

Firm Responses to Clean Energy Tax Credit Transfers: Investment, Tax Planning, and Disclosures*

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Abstract: Clean energy tax credits are a primary way for governments to motivate firms to invest in renewable energy. I study firm responses to a recent reform that dramatically increased access to clean energy tax credits. Specifically, I examine the transferability of clean energy tax credits, introduced with the Inflation Reduction Act (IRA) of 2022, which allows clean energy firms to sell credits for cash. This created a new market, yet little is known about how firms respond to it. Using a novel hand-collected dataset of disclosed tax credit transfers from 2023 through 2025, I examine firm responses by buyers and sellers. On the seller side, I use stacked cohort difference-in-differences designs to examine whether improved credit monetization affects clean energy investment and investment quality. I find that selling firms increase capital expenditures, productive assets, and generators after their first transfer, with the increase concentrated in green energy investment. On the buyer side, participation expands sharply and diversifies relative to the pre-IRA tax-equity market. Buying firms are more profitable, have higher ESG scores, and are more likely to have emissions targets than other potential buyers. I further examine whether tax credit purchases affect buyers' effective tax rates (ETRs) and how firms discuss these transactions in their sustainability reports. I find no significant decline in ETRs following a purchase, but that firms incorporate tax credit purchases into their sustainability reports as part of broader environmental narratives. Overall, my findings show that tax credit monetization affects real investment and corporate participation in clean energy finance.

Keywords: Clean energy tax credits, Transferability, Firm Responses

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1 Introduction

Firms worldwide face societal and political pressure to reduce greenhouse gas emissions. In response to the growing economic and environmental consequences of climate change, governments have adopted increasingly ambitious emissions-reduction commitments and net-zero targets. Meeting these commitments requires large-scale private sector investment in clean energy, as energy production accounts for the majority of global greenhouse gas emissions (International Energy Agency 2024a). Such investment, however, is highly capital intensive, and the high upfront costs of capital are consistently identified as a major barrier to the energy transition by academics (e.g., De Haas et al. 2025), industry bodies (International Energy Agency 2024b), and government agencies (Intergovernmental Panel on Climate Change 2023; European Commission 2025).¹ To lower the costs of clean energy investment and encourage firms to invest, governments rely on clean energy tax credits, which reduce the taxes a firm owes when it invests in or produces clean energy (Metcalf 2010). In this paper, I study how firms respond to clean energy tax credits.

Tax credits reduce firms' tax liabilities. However, many clean energy developers have little taxable income during development and thus cannot immediately use the credits, limiting the value of the tax incentive. Before 2023, a developer that could not use its credits had one main way to turn them into cash, a relatively complex arrangement known as tax-equity financing (Garrett and Shive 2025).² Access to tax-equity financing was concentrated, and developers often realized substantially less than the full value of their tax benefits. Many developers therefore remained unable to readily monetize their tax credits, raising concerns about the effectiveness of these incentives.³ The U.S. Inflation Reduction Act (IRA) of 2022 dramatically changed this by making clean energy tax credits transferable. This means a developer can now sell a credit it cannot use to an unrelated

¹Global estimates of the required annual investment range from \$3.5 to 9.2 trillion (World Economic Forum 2023).

²The developer partnered with an outside investor that owed substantial taxes. The investor contributed capital to the project in exchange for the credits the developer generated, which it used to reduce its own tax bill. These partnerships were legally complex and expensive to arrange, and participation was dominated by a small group of large banks and insurers with the tax capacity and expertise to structure them.

³For example, in testimony before the House Energy and Commerce Committee's Subcommittee on Energy, Craig Gordon of Invenergy, a leading clean energy developer in the United States, emphasized that the supply of tax equity has always been limited and highly concentrated" and that only a limited amount of tax equity was available for renewable projects (U.S. House Committee on Energy and Commerce, Subcommittee on Energy 2021). In separate testimony before the House Select Committee on the Climate Crisis, John Larsen of the independent research organization Rhodium Group described the tax-equity market as having hit its functional limits" and reported that developers were either unable to obtain financing or faced very high financing costs because of these constraints (U.S. House Select Committee on the Climate Crisis 2021).

firm, receiving cash immediately. In return, the buyer acquires the credit at a discount and applies it against its own taxes. By removing the need for a tax-equity partnership and creating a new market for tax credits, transferability is intended to broaden access to clean energy financing beyond the large institutions that dominated tax-equity.

This change represents a major shift in the market. For example, former Treasury Secretary Janet L. Yellen, described the new monetization tool as a “force multiplier, enabling companies to realize far greater value” from clean energy incentives. White House National Economic Advisor Lael Brainard stated that transferability lets “more small businesses, startups, and other businesses . . . benefit from game-changing clean energy tax credits” (U.S. Department of the Treasury 2024). Reflecting this substantial shift, transfer volume reached roughly \$ 24 billion in 2024, surpassing the size of the tax-equity market (\$20 billion transfer volume annually) in only its second year. In the next year, the volume *doubled* to \$42 billion (Crux 2025).⁴ My data on the number of firms participating in the market confirms this substantial increase (see Figure 1).

Despite the scale and growth of this market, little is known about which firms participate in transfer tax credits and how transferability affects clean energy investment. In this paper, I examine the effects on both the developers that sell tax credits (*sellers*) and the corporations that buy them (*buyers*). Understanding these responses is important because there is an active debate among policymakers on the future of the underlying tax credits. The One Big Beautiful Bill Act of 2025 accelerated the phase-out of the underlying credits, but policymakers from both parties have since introduced bills to restore them.⁵ Evidence on how firms respond to transferability directly informs these proposals.

On the seller side, I test two hypotheses. I study whether the ability to sell tax credits i) increases clean energy investment, ii) if so, whether it affects the quality of that investment. I first consider the effect on investment levels. Transferability can affect investment through two channels. The first is a monetization channel. Firms with limited tax liability cannot fully realize the value of the credits they generate. Allowing these firms to sell their credits for cash raises the realized value of the tax incentive, lowers the after-tax cost of eligible investment, and can make additional projects profitable (Hall and Jorgenson 1967). The second is a financing channel. The cash proceeds increase internal funds and reduce firms’ reliance on external financing, which can increase investment, particularly among financially constrained firms (De Haas et al. 2025; Blonz et al. 2026).

⁴By further comparison, the market for transferable credits now approaches the size of the entire federal research and development credit, which totaled roughly \$43 billion in 2022 (Internal Revenue Service 2025).

⁵See the Energy Bills Relief Act, introduced in the House by Democrats in March 2026, and the American Energy Dominance Act, introduced in the House by Republicans in April 2026.

Although both channels predict a positive investment response, the magnitude of the effect is uncertain. This is because tax-equity financing already allowed some firms to monetize their credits and because transaction costs, buyer demand, and uncertainty in the new transfer market may limit the value firms realize (Gallemore et al. 2025; Garrett and Shive 2025).

I next turn to my second hypothesis, which examines whether transferability affects investment quality. The same frictions that affect whether a project is undertaken and the level of investment can also affect which projects firms undertake and how well those projects perform. By alleviating monetization and financing frictions, transferability can increase the internal funds available to develop, complete, and operate generating assets, potentially improving investment quality. At the same time, transferability may reduce quality for at least two reasons. First, the marginal projects made profitable by the reform may have lower expected operating productivity (Eichfelder et al. 2023; Chan and Fischer 2026). Second, to the extent that transferability substitutes for tax-equity financing, it may weaken the monitoring provided by investors with an ownership stake in the project (Garrett and Shive 2025). These competing predictions make the effect of transferability on the realized operating performance of clean energy investment ambiguous *ex ante*.

To study the responses to tax credit transferability, I hand-collect a novel dataset of clean energy tax credit transfers. Because no comprehensive data source identifies these transactions, I assemble the data from SEC filings and public announcements, identifying 218 firms that participate as buyers or sellers between 2023 and 2025. For each transfer, I record the disclosed parties and, when available, the credit type, amount transferred, and transaction price.⁶ To my knowledge, this is the first systematic firm-level dataset to identify participants on both sides of the transferable tax credit market.

I first test my first hypothesis on whether transferability increases investment on the seller side. To estimate these effects, I link the hand-collected sample of electric utilities that sell tax credits to financial statement data from Compustat. I then use a stacked cohort difference-in-differences design that compares firms around the year of their first tax credit transfer with otherwise eligible firms that do not transfer a credit. I find that firms increase investment after entering the transfer market. In my preferred specification, capital expenditures increase by 1.2 percentage points of beginning total assets, or approximately 20 percent relative to the sample mean, while fixed assets relative to beginning total assets increases by 6.9 percentage points, or approximately 7 percent relative to

⁶I observe a transfer only when at least one party publicly discloses it. I benchmark the aggregate volume identified in my sample against external estimates of the transfer market and confirm that the sample captures a substantial share of estimated market volume over the sample period.

the sample mean.

In addition to testing investment with financial data, I complement my sample with plant-level data from the U.S. Energy Information Administration (EIA). This allows me to observe a firm's number of electrical generators, which are the predominant asset purchased by clean energy developers. When using this more discrete measure of equipment, I find that firms add approximately four generators. I then use the granularity of the EIA data to classify generators as clean or non-clean and examine which type of investment increases. The increase in generators is concentrated in clean generators, with no corresponding increase in non-clean generators. This result indicates that transferability increases not only overall investment but specifically investment in the clean generating assets targeted by the policy.

To examine the channels underlying the investment response, I next use cross-sectional tests to assess whether I next use cross-sectional tests to examine whether the investment response is consistent with the monetization or financing channel developed above or both. The monetization friction should be more binding for firms with limited tax capacity because these firms have greater difficulty using the credits they generate directly. Consistent with this prediction, investment responses are stronger among firms with lower profitability, greater net operating losses, and pretax losses. The financing friction should be more binding for firms with more limited access to external capital. Across several measures of financing constraints, I find patterns consistent with stronger investment responses among more financially constrained firms. Together, these results provide evidence consistent with both channels through which transferability can increase investment.

Finally, I examine whether the response varies with firms' existing exposure to clean energy. This distinction matters because transferability may either reinforce investment by firms already active in clean energy or encourage firms with less clean generating portfolios to expand into clean energy. I find stronger responses among firms with a lower pretreatment share of clean generation. This finding suggests that transferability expands clean energy investment beyond firms that were already relatively clean and facilitates clean energy expansion among firms with greater scope to shift their generating portfolios toward clean energy.

I then turn to my second hypothesis and examine whether the additional investment is accompanied by changes in investment quality. Another benefit of the EIA data is that it also provides investment quality measures, such as generator capacity factors and average generation per generator, which allow me to examine whether the additional investment is accompanied by changes in realized operating performance. I find no evidence that tax credit transfers affect electricity generation relative to available generating capacity or av-

erage generation per generator. I similarly find no consistent effect on pretax ROA, which I use as a supplementary firm-level performance measure. Thus, while transferability increases the amount of clean energy investment, I find no corresponding improvement or deterioration in the average operating performance of firms' generating assets over the period I observe. This means that transferability expands clean energy investment without evidence that the additional investment is either higher or lower quality on average.

In the second part of the paper, I turn to the buyer side, providing what is, to my knowledge, the first systematic evidence on which firms purchase clean energy tax credits and what predicts their participation in this new market. Before transferability, the tax-equity market was dominated by large banks and other financial institutions because participation required substantial taxable income and involved high transaction costs. I first examine whether transferability broadened participation beyond these traditional tax-equity investors. I compare tax credit buyers with the tax-equity investors that financed clean energy projects before the IRA and find that participation expands sharply after transferability is introduced. Whereas new tax-equity partnerships ranged from 5 to 12 per year before the IRA, the number of firms purchasing tax credits exceeds 50 in 2024 and reaches roughly 100 in 2025. The firms entering this new market are economically significant, with publicly traded buyers having an average market capitalization of \$25.4 billion. The composition of participants also broadens substantially. Investors in the earlier tax-equity market were concentrated almost entirely in the finance and insurance industries, while tax credit buyers span 13 additional industries, including manufacturing and retail.⁷

I then compare tax credit buyers with the broader set of corporations that could purchase credits and descriptively examine which pre-market characteristics predict participation, focusing on firms' ability to use tax credits, tax sophistication, environmental orientation, and other firm characteristics. Descriptive comparisons show that, before the market opened, future tax credit buyers had higher pretax income and paid more cash taxes relative to assets. They also had higher ESG scores and were more likely to have committed to emissions targets. These patterns suggest that having sufficient taxable income and tax payments to absorb the credits is an important factor in participation, while environmental orientation is also associated with firms' willingness to enter the market.

I next examine what firms gain from purchasing tax credits by studying their effective tax rates (ETRs) and sustainability disclosures. The most direct economic benefit arises because buyers purchase credits for less than their face value and apply the full credit amount

⁷Examples of tax credit buyers include Keurig Dr Pepper, Tripadvisor, Best Buy Co., and McDonald's Corporation.

against their tax liabilities. The difference between the credit's face value and its purchase price therefore represents a direct economic benefit to the buyer and provides a new tax planning opportunity without requiring ownership of the underlying clean energy project. I examine whether this benefit is reflected in firms' ETRs around their first purchase. I find no evidence of a significant decrease in ETRs after buying a tax credit. This lack of a detectable effect may reflect differences in the timing of credit purchases and use, as well as the fact that ETRs capture firms' overall tax positions and other tax planning activities, making the incremental effect of a credit purchase difficult to isolate.

To examine whether tax credit purchases provide benefits beyond direct tax savings, I also study how firms disclose their purchases in sustainability reports. A tax credit purchase can link a buyer to clean energy investment and provide content that can be incorporated into broader sustainability narratives (Lin et al. 2024). Because the buyer does not own or operate the underlying project, however, the connection between the purchase and the environmental benefits generated by the project, such as emissions reductions, is indirect. This creates scope for firms to present tax credit purchases as evidence of environmental commitment even when the transaction has a limited connection to the firm's own environmental impact, raising concerns related to greenwashing (Heath et al. 2023; Bilicka et al. 2025; Duchin et al. 2025). To examine how firms present these transactions, I collect and review the sustainability reports of tax credit buyers. My exploratory review shows that approximately half of the buying firms that publish a sustainability report emphasize the connection between their tax credit purchases and clean energy investment or incorporate the purchases into broader sustainability narratives.⁸

This paper makes several contributions. First, the paper speaks to the literature on the real effects of environmental tax policy (Lester and Olbert 2025). One strand of this literature studies carbon taxes and examines their effects on capital investment (Jacob and Zerwer 2024), R&D spending (Brown et al. 2022), firm emissions (Martinsson et al. 2024; Colmer et al. 2025), and carbon leakage (Känzig et al. 2024). In contrast, a related strand of concurrent work examines green investment incentives, including the effects of investment tax credits on electric utilities' generation and emissions (Chan and Fischer 2026) and the effects of government loans and subsidies on corporate environmental performance (Cincotta and Koester 2026). I extend this literature by studying a distinct margin of environmental policy in which transferability changes firms' ability to realize the value of an existing tax incentive rather than whether the incentive is available or how large it is. My findings show that the effectiveness of clean energy tax incentives depends not

⁸For example, one corporate buyer describes its purchase of renewable energy tax credits in its CSR report as a way "to further bolster our own commitment to the energy transition."

only on their statutory value, but also on firms' ability to monetize them and finance the underlying clean energy investment.

Second, to my knowledge, I provide the first systematic evidence on the corporate buyers of transferable clean energy tax credits. Prior work on clean energy tax incentives and tax-equity financing primarily examines the firms and projects receiving the incentives or financing (Garrett and Shive 2025; Blonz et al. 2026; Chan and Fischer 2026). I complement this literature by examining the demand side of the new market. I find that buyer participation expands substantially relative to the pre-IRA tax-equity market and extends beyond the financial institutions that traditionally supplied tax-equity capital. These findings contribute to the clean energy financing literature by showing how a broader corporate buyer base supports the monetization of tax incentives and the flow of capital to clean energy investment (De Haas et al. 2025; Lanteri and Rampini 2025).

Finally, the paper provides evidence relevant to the ongoing policy debate over clean energy tax incentives. Transferability created a rapidly growing market while at the same time policymakers are reconsidering the underlying credits as a policy tool to encourage green investment. My findings show how easier monetization increases clean energy investment and participation in clean energy finance. More broadly, the results highlight that the effectiveness of a tax incentive depends not only on its statutory value, but also on firms' ability to monetize it.

2 The New Market for Tax Credits

2.1 Institutional Details

Clean energy tax credits reward firms for investing in or producing clean energy. The two principal credits are the Investment Tax Credit under Sections 48 and 48E, which is based on a project's eligible investment cost and earned when the project is placed in service, and the Production Tax Credit under Sections 45 and 45Y, which is based on electricity generated and earned over the years a project operates.⁹ The Inflation Reduction Act of 2022 expanded existing clean energy credits and introduced new incentives for technologies such as clean hydrogen, advanced manufacturing, carbon capture, and clean fuels.¹⁰ For taxable firms, these credits reduce federal income tax liability dollar-for-dollar, thereby lowering the after-tax cost of clean energy investment. Figure 2 Panel (a) illus-

⁹For facilities placed in service after 2024, Sections 48E and 45Y replace Sections 48 and 45. The transition rules allow certain projects that began construction before 2025 to remain eligible for Sections 48 and 45.

¹⁰Appendix Table A.1 lists the eligible credits, their statutory sections, and the technologies they cover.

trates this mechanism for a clean energy firm. The firm earns a tax credit by undertaking an eligible investment or producing eligible clean energy and claims the credit against its federal tax liability. The credit reduces the amount of tax the firm owes by its allowable value. The value of the incentive to the firm, however, depends on its ability to use the credit. A firm with insufficient tax liability may be unable to realize the full value of the credit in the year it is generated.

This tax capacity requirement creates a central friction for clean energy firms. Many developers are young, capital-intensive, and report little taxable income in the years they generate credits, limiting their ability to use the credits immediately. Before the IRA, the primary way to monetize credits that a firm could not readily use was tax-equity financing. Figure 2 Panel (b) illustrates a typical tax-equity transaction. In this structure, a clean energy firm partners with an investor that has substantial tax liability. The investor contributes capital to the project in exchange for tax credits and other tax benefits. These arrangements are complex and costly and require substantial legal, tax, and accounting expertise. As a result, tax-equity investment has historically been concentrated among a relatively small group of investors, particularly banks, insurers, and large corporations. As a result, many clean energy firms faced limited access to this source of financing. In addition, the legal, administrative, and compliance costs associated with tax-equity structures can reduce the value developers realized from the underlying tax benefits by an estimated 20 to 30 percent (U.S. Government Accountability Office 2015; Congressional Research Service 2019).¹¹

Section 6418 of the IRA introduced transferability as an alternative.¹² Figure 2 Panel (c) illustrates this mechanism. A clean energy firm that earns an eligible credit can elect to sell all or part of it to an unrelated buyer in exchange for cash.¹³ The buyer pays cash, which the seller does not include in taxable income, and applies the credit against its own tax liability. For applicable credits, a buyer that cannot use the full credit in the year of purchase can carry the unused portion back up to three years or forward up to twenty

¹¹Tax-equity transactions also monetize accelerated depreciation in addition to the credits, a benefit transferability does not provide. Hybrid structures that combine a tax-equity partnership with a transfer have become common.

¹²Policymakers favored transferability over direct subsidies or expanded direct-payment programs for several reasons. The IRA passed through budget reconciliation, which limited new direct spending, and a transferable credit is scored as forgone revenue rather than government spending, thereby preserving the political framing of the support as a tax provision. Transferability also shifts the task of evaluating and pricing project risk to private market participants rather than the government, which is intended to limit fraud and improve project selection.

¹³The seller must register the eligible credit property with the IRS and obtain a registration number. The seller and buyer then complete a transfer election statement, which each attaches to its tax return. The statement must be completed before either party files its relevant return.

years.¹⁴ The price is below the credit's face value, so the seller receives less than full value in exchange for immediate liquidity, while the buyer reduces its tax liability by more than the amount it pays. In 2024, production tax credits traded at an average of 95 cents per dollar of credit and investment tax credits at 92.5 cents (Crux 2025), allowing sellers to realize substantially more of the credit's face value than under traditional tax-equity financing. The discount reflects transaction costs, tax risk, and market conditions. Unlike tax-equity financing, a direct transfer does not require the buyer to take an ownership stake in or participate in the operation of the underlying project. Although the reform does not require an intermediary, many transfers are arranged through platforms and brokers that match buyers and sellers and support the transaction.¹⁵

Two features of transferability are important for the analysis that follows. First, the credit can be separated from ownership of the underlying project. The buyer has no operational or governance rights and does not receive an ownership interest or claim on the project's operating cash flows. Second, the buyer nonetheless bears the tax risk associated with the credit. For credits subject to recapture, the buyer is liable for any recapture attributable to the portion it purchased, even though it does not control the project. For example, an investment tax credit can be recaptured during the five years after the project is placed in service if the underlying property is sold or otherwise ceases to qualify including when financial distress or insolvency results in a disposition of the project.¹⁶

The IRA also restructured the amount of the credit available for transfer. For many clean energy credits, taxpayers receive a base amount that can be substantially increased through bonus provisions. Most importantly, compliance with prevailing-wage and apprenticeship requirements increases the base credit amount fivefold.¹⁷ Certain credits also provide additional bonuses for domestic content and projects located in energy or low-income communities.¹⁸ As a result, the dollar value of an earned credit can vary across otherwise similar projects. These provisions affect the amount of the credit generated and therefore the amount available for transfer, but not the transfer mechanism itself. Once the eligible credit amount is determined, including any applicable bonus amounts, the seller can transfer all or a proportionate portion of the credit.

¹⁴The credit remains subject to the general business credit limitation under Section 38. Any unused portion of an applicable credit can be carried back up to three years, beginning with the earliest year, and forward up to twenty years.

¹⁵Platforms such as Crux and Reunion match counterparties and assist with pricing, due diligence, documentation, and compliance.

¹⁶The seller is responsible for substantiating the underlying credit and must notify the buyer of a recapture event. Buyers can obtain insurance against recapture risk mitigating the economic risk.

¹⁷For example, the Section 48E investment credit increases from 6 percent to 30 percent of qualified investment when these requirements are satisfied.

¹⁸Appendix X summarizes these requirements. **Need to be added.**

The One Big Beautiful Bill Act of 2025 accelerated the termination of several clean energy credits, shortening the period during which new projects can qualify. These changes were enacted in July 2025, during the final year of my sample, and primarily affect the eligibility of future projects rather than the transfer mechanism that I study.

2.2 Identifying Tax Credit Transfers

Because tax credit transfers are new, no existing dataset identifies firm level tax credit transfers. I therefore hand-collect transfers from public disclosures. I first use the SEC EDGAR full-text search to identify public firms that mention tax credit transfers or related terms. For each identified firm, I then review its filings covering 2023 through 2025, including 10-Ks, 10-Qs, 8-Ks, and relevant exhibits, to identify the first year in which it sells or purchases a credit.¹⁹ I supplement these filings with news coverage from Factiva and targeted Google searches. These additional sources allow me to identify transfers involving private firms and to reconcile transaction details across multiple sources. Across the 2023 to 2025 treatment window, I identify 80 firms that sell a transferable credit and 138 firms that purchase one.²⁰ Where disclosed, I record the credit amount, transaction price, transaction costs, and credit type, although these details are not reported consistently across firms. I use the first observed year in which a firm sells or purchases a transferable credit to define participation and treatment timing.

Because not every transfer is disclosed, I assess how much of the market my hand-collected sample captures. Appendix Figure A.1 Panel (a) compares the aggregate transfer volume by year in my sample against the market estimates reported by Crux, a major tax credit transfer platform that facilitates transfers.²¹ My sample captures roughly 70 percent of estimated market volume in 2023, declining to about half by 2025. These figures indicate that my sample captures a substantial share of the dollars transferred in a market for which no complete public record exists.²² The lower coverage in later years is consistent with the market expanding to private developers and smaller transactions that

¹⁹I observe a transfer only if at least one party publicly discloses it. Because the amounts involved are material, public firms typically report them, so the sample should capture the economically significant activity in the market. I assess the coverage of my hand-collected sample below.

²⁰A single transaction has both a buyer and a seller, which I record separately. I often observe only one party to a transaction and therefore cannot match most buyers and sellers. The larger number of observed buyers is consistent with many clean energy developers being private and therefore having fewer public disclosure requirements. To the extent that I fail to identify firms that sell or purchase tax credits, these firms may be misclassified as controls. Such misclassification would make it more difficult to detect differences between treated and comparison firms.

²¹For transfers that do not disclose a transaction price, I impute the price using the mean within credit type and technology group.

²²The Crux market totals are themselves estimates rather than observed population totals.

are less likely to appear in public disclosures. Reporting lags may also affect 2025 because some transfers for the 2025 tax year are first disclosed in filings made during 2026. The median price in my sample is 93 cents per dollar of credit, close to market prices reported by Crux (2025). The average transfer size is \$208 million. **CHECK Numbers again.**

Appendix Figure A.1 Panel (b) plots the number of observed sales and purchases by year. Market participation grows rapidly following the first transfers in 2023. I record 17 credit sales in 2023 and 42 in 2025, while the number of purchases increases from 12 to 105. Investment tax credits are the most frequently transferred credits in the sample, followed by production tax credits.

3 Hypothesis Development Seller-Side

I first ask whether transferability affects a firm's clean energy investment. In the neoclassical user cost of capital framework of Hall and Jorgenson (1967), an investment credit of k per dollar of investment reduces the price of new capital, lowering the user cost of capital and raising a firm's desired capital stock.²³ A tax credit therefore reduces the after-tax cost of investing in eligible assets and should lead firms to invest more. This framework assumes no frictions. In such a frictionless setting, the tax credit would be worth its face value regardless of whether the firm uses it directly, monetizes it through a tax equity partnership, or transfers it for cash. Transferability would therefore leave the user cost unchanged and have no additional effect on investment. In practice, however, two frictions can prevent clean energy firms from realizing the full value of the credit.

The first is a monetization friction. Clean energy credits are nonrefundable, so a firm with little or no tax liability cannot apply a credit against its own taxes and thus cannot realize its value directly.²⁴ Firms with access to tax-equity financing can monetize the credit indirectly, but these arrangements are costly to structure and reduce the value ultimately realized by the firm. Thus, whether a firm lacks sufficient tax liability or relies on costly tax-equity financing, the effective value of the credit entering the user cost of capital can fall below its statutory credit amount k . As a result, some projects that would have been profitable if the firm could realize the credit's full value no longer clear the investment hurdle. By allowing the firm to sell the credit for cash at a relatively small discount to face

²³A standard representation of the user cost of capital is $c = q(r + \delta) \frac{1-k-uz}{1-u}$, where q is the price of capital, r is the cost of funds, δ is economic depreciation, u is the corporate tax rate, and z is the present value of depreciation deductions per dollar of investment.

²⁴Unused credits can generally be carried to other tax years, but this delays realization of the tax benefit and can reduce its present value, particularly for early-stage firms and projects that may take several years to generate sufficient taxable income.

value, transferability raises the realized value of the credit closer to k . This lowers the user cost of capital and increases the after-tax return on investment. Transferability therefore attenuates the monetization friction, making additional eligible projects profitable and increasing the firm's investment.

The second is a financing friction. Clean energy generation is capital intensive and slow to generate revenue, and a large share of the project costs are incurred upfront (Blonz et al. 2026). Whether a project proceeds therefore often depends on a firm's ability to raise sufficient external financing, even when the project has positive net present value. When a firm transfers a credit, it converts a tax benefit that may otherwise remain unusable into an immediate cash inflow. These proceeds increase the firm's internal funds and can be reinvested in existing or new clean energy projects, reducing the amount of external financing the firm must raise (De Haas et al. 2025). Whereas the monetization friction operates through the effective value of the credit k , the financing friction operates through r , the cost of funds in the user cost of capital. By increasing internal liquidity and reducing reliance on external finance, transferability can lower the effective financing costs associated with investment and therefore reduce r . Consistent with this mechanism, Blonz et al. (2026) find that smaller firms experience a decline in borrowing costs relative to larger firms following the IRA and argue that these firms are more likely to benefit from credit transferability. At the margin, improved access to financing can allow a constrained firm to undertake a project it would otherwise defer or forgo. Transferability therefore attenuates the financing friction by providing cash that can be reinvested, with the investment response expected to be stronger among financially constrained firms.

These two frictions are distinct but can reinforce one another. The monetization friction concerns whether a firm can realize the value of a tax credit, while the financing friction concerns whether it has sufficient funds to undertake the investment. The two can arise together because firms with little taxable income may also face greater financing constraints (Blonz et al. 2026). The effect of transferability therefore depends partly on the extent to which tax-equity financing already alleviated these frictions. Firms with access to tax equity could monetize their credits and obtain project financing before the reform, but only through relatively costly and complex arrangements. Transferability therefore represents a larger change for firms that previously lacked access to tax equity, while also benefiting firms that already used tax equity by allowing them to monetize credits through a simpler and potentially less costly structure.

It is also unclear ex ante how effectively the new transfer market can alleviate these frictions. Finding a buyer, pricing the credit, conducting due diligence, and completing the transaction impose costs that reduce the net value realized by the seller. Transfer-

ability also requires sufficient demand from firms with enough tax liability to absorb the credits offered for sale. If transaction costs are high or buyer demand is limited, transferability may provide little improvement over existing monetization options. Uncertainty about the new market and the durability of the underlying tax incentives may further limit firms' responses. Thus, although both mechanisms predict that transferability can increase investment, its effect is uncertain because it depends on how binding the underlying frictions are and how effectively the new market alleviates them. Given this ambiguity, I state the hypothesis in the null.

H1: *Tax credit transferability is not associated with the level of clean energy investment.*

The preceding discussion considers whether transferability increases the amount firms invest. An increase in investment, however, does not necessarily imply an increase in investment quality, which I define as the realized operating performance of a firm's investments. The same frictions that affect whether a project is undertaken can also affect which projects firms undertake and how well those projects perform. I therefore next ask whether transferability affects the quality of clean energy investment.

On the one hand, alleviating monetization and financing frictions can improve investment quality. First, reducing the monetization friction increases the value firms realize from their tax credits, providing additional resources that can be used to develop, complete, and operate clean energy projects. If imperfect credit monetization otherwise limits the resources available for productive investment, transferability can improve the realized performance of those projects. Second, alleviating the financing friction can have a similar effect. A financially constrained firm may not only defer otherwise profitable projects but may also lack the resources to fully fund their development and operation. By converting tax credits into cash, transferability increases internal funds that can be used to complete existing projects, maintain generating assets, or devote additional resources to project development and operation (Goolsbee 2004). If financing constraints otherwise lead firms to scale back or delay productive investment, relaxing these constraints can improve the performance of the projects firms undertake. Both channels therefore predict that transferability can increase investment quality alongside investment levels.

On the other hand, transferability can reduce the average quality of investment through two channels. First, by increasing the value firms realize from tax credits and lowering the effective cost of capital, transferability makes additional projects profitable. These marginal projects may have lower expected operating productivity than the projects firms would have undertaken absent transferability.²⁵ Thus, transferability can change

²⁵A related possibility is that the broader transfer market attracts buyers with less experience evaluating

not only how much firms invest but also the composition of projects that firms undertake. Consistent with this possibility, Eichfelder et al. (2023) bonus depreciation incentives for business investment in East Germany and find that tax incentives increase investment while reducing the ability of current investment to generate future revenues. Similarly, Chan and Fischer (2026) examine investment tax credits for electric utilities in the United States and find that electric utilities increase investment following the adoption of investment tax credits but experience a smaller increase in subsequent electricity generation. These findings suggest that tax incentives can increase investment without producing a corresponding increase in operating performance. Transferability may therefore increase investment while lowering the average performance of the projects undertaken.

Second, transferability changes the monitoring associated with project financing. Under tax-equity financing, the investor obtains an ownership interest in the project and has incentives to conduct due diligence and monitor its development and operation. A purchaser of a transferable credit instead acquires the tax benefit without obtaining an ownership interest or operational control. Projects financed through transfers may therefore receive less external monitoring than projects financed through tax equity. This difference can matter both when transfers substitute directly for tax equity and when transferability enables projects that otherwise might have required tax-equity financing to proceed. Garrett and Shive (2025) find that wind projects financed by tax-equity investors produce more electricity relative to their generating capacity and attribute this difference to monitoring and contracting rather than project selection. If transfer financing provides less comparable oversight, the operating performance of the projects undertaken may be lower.

Taken together, transferability can increase investment quality by alleviating monetization and financing frictions and providing firms with additional resources to develop and operate their projects, but it can also reduce average project quality by making lower-productivity marginal projects profitable and weakening the monitoring associated with tax-equity financing. The net effect on investment quality is therefore ambiguous ex ante. I therefore state the hypothesis in the null.

H2: *Tax credit transferability is not associated with the quality of clean energy investment.*

clean energy tax credits and the underlying projects. If these buyers conduct less effective screening, lower quality projects may face fewer barriers to obtaining financing. I examine which firms enter the buyer side of the transfer market in Section 6.1.

4 Research Design and Data Seller-Side

4.1 Research Design

To test [H1](#) and [H2](#), I define treated firms as firms that sell a tax credit and classify them as treated beginning in the year of their first observed transfer. Ideally, to estimate the effect of transferability on investment, I would compare a treated firm’s investment after its first transfer with the investment the same firm would have undertaken in the absence of the transfer. Because this counterfactual is unobservable, I use firms that operate in the same clean energy markets, could plausibly generate eligible tax credits, and do not make an observed transfer during the sample period as the comparison group.

Firms first enter the transfer market in different years (2023, 2024, or 2025 in my sample). I therefore use a stacked cohort DiD design. For each treatment cohort, I construct a separate sample that compares firms first treated in that year with eligible firms that remain untreated during the relevant cohort window. I align each cohort around its first treatment year and combine the cohort-specific samples for estimation. This approach prevents firms that have already transferred a credit from serving as controls for firms treated later and avoids problematic comparisons that can arise in conventional two-way fixed effects models when treatment effects vary across cohorts or over time (Baker et al. [2022](#); Baker et al. [2026](#)).

I measure the level of investment using three primary outcomes. $CapExInvestment_{i,t}$ equals capital expenditures divided by total assets at the end of year $t - 1$. This measure captures investment intensity relative to the firm’s existing asset base. $PPEInvestment_{i,t}$ equals gross property, plant, and equipment divided by total assets at the end of year $t - 1$ and captures the firm’s stock of productive assets relative to its existing asset base. $Generators_{i,t}$ measures the number of distinct electricity generating units operated by firm i in year t . I use $\log(Generators_{i,t} + 1)$ in the primary specification. The first two measures capture capital spending and the stock of productive assets using financial statement data, while the number of generators provides a more granular physical measure of electric utilities’ generating assets. Together, the measures capture different but related dimensions of investment, providing complementary evidence on firms’ investment responses. As robustness tests, I also examine the logarithm of capital expenditures and gross property, plant, and equipment and estimate a PPML specification for the number of generators (Santos Silva and Tenreiro [2006](#); Cohn et al. [2022](#); Chen and Roth [2024](#)).

To test [H2](#), I measure investment quality using the realized operating performance of firms’ generating assets. My primary measure is the capacity factor, which measures actual

electricity generation relative to the amount that could have been generated if the firm's installed capacity operated continuously at full capacity. A higher capacity factor therefore indicates greater realized electricity generation relative to the maximum potential output of the firm's installed capacity. I also examine average electricity generation per generating unit as an alternative measure of operating performance. Together, these outcomes allow me to assess whether the additional generating assets associated with tax credit transfers are accompanied by changes in their realized operating performance. As supplementary evidence, I examine pretax return on assets, calculated as pretax income divided by beginning total assets. Because ROA reflects the performance of the entire firm rather than its generating assets, I interpret it as a complementary rather than primary measure of investment quality.

I estimate the following model:

$$Y_{i,c,t} = \beta_1 \times (Treat_{i,c} \times Post_{c,t}) + \gamma' X_{i,c,t-1} + \lambda_{i,c} + \omega_{t,c} + \varepsilon_{i,c,t} \quad (1)$$

where $Y_{i,c,t}$ is an investment level or quality outcome for firm i in cohort stack c and year t . $Treat_{i,c}$ equals one for firms first treated in cohort c , and $Post_{c,t}$ equals one in the treatment year and subsequent years. $\mathbf{X}_{i,c,t-1}$ is a vector of one-year-lagged firm characteristics. For the financial statement analyses, the controls include log total assets, ROA, and leverage, while for the electricity generation analyses they include log net generation and the percentage of generation from clean sources. Firm by cohort fixed effects $\lambda_{i,c}$ absorb time-invariant differences between treated and control firms within each cohort. Year by cohort fixed effects $\omega_{t,c}$ absorb common shocks within each cohort-specific comparison. The coefficient of interest is β_1 , which captures the change in the investment outcome following a firm's first tax credit transfer relative to the change among eligible untreated firms. As a robustness test, I replace the lagged controls with pre-treatment averages of the same firm characteristics interacted with year indicators. This alternative specification avoids conditioning on characteristics that may themselves respond to an earlier tax credit transfer while allowing firms with different pre-treatment characteristics to follow different trends over time.

Two limitations are inherent to the setting. First, firms choose whether to transfer their tax credits, raising concerns that treated and control firms differ systematically before treatment. Firm fixed effects absorb time-invariant differences across firms, while the lagged controls account for differences in observable firm characteristics over time. To further improve comparability between treated and control firms before treatment, I apply entropy balancing separately within each treatment cohort (Hainmueller 2012). I reweight the comparison firms so that the pre-treatment means of the balancing variables

match those of the treated firms. I balance on pre-treatment values of the same financial and operating characteristics used as controls, including firm size, ROA, leverage, net generation, and the share of clean generation.²⁶ Entropy balancing does not eliminate selection on unobservable characteristics, but it reduces observable differences between firms that do and do not transfer credits.

A second limitation is the short history of the transfer market. The 2023 treatment cohort is observed for three post-treatment years, while the 2024 and 2025 cohorts are observed for two and one post-treatment years, respectively. The main specification therefore captures effects over the post periods currently observable. As a robustness test, I separately estimate the specification for firms first treated in 2023, which provides three pretreatment and three post treatment years around the first transfer, and report a column restricted to this cohort directly within the main tables below, with the corresponding event-study evidence in Appendix Table B.5 and Figure B.2.

4.2 Data and Sample Construction

Table 1 reports the construction of the seller-side sample. I begin with the hand-collected sample of firms that disclose selling a transferable tax credit between 2023 and 2025. This procedure identifies 80 seller firms and 240 seller firm-year observations. I classify these firms as electric utility or non-utility firms. My main analysis focuses on firms that generate electricity because they represent the largest group of treated firms and provide the most comprehensive data for measuring investment outcomes.²⁷ To study the electric utility firms, I merge the tax credit sellers with plant and generation data from the Energy Information Administration. I use Forms EIA-860 and EIA-923 covering 2018 through 2025 and keep eligible electric utilities that do not match to an observed seller as the comparison group. I exclude tax-exempt municipal utilities, cooperatives, and public power entities because they do not provide a comparable counterfactual for taxable firms using transferability. I also require sufficient data coverage around the treatment period and non-missing data to construct the control variables.²⁸ These restrictions result

²⁶Appendix Table B.3 compares pre-treatment characteristics across treated firms, the unweighted baseline control group, and the entropy-balanced control group. Entropy balancing eliminates differences in the matched means by construction. I do not also balance on second moments because of the small number of firms within some treatment cohorts. Adding a second-moment constraint for each covariate can result in concentrated weights being placed on a small number of comparison firms.

²⁷I'm planning to examine non-utility sellers separately. These firms include manufacturers of solar panels and wind tower components as well as clean fuel producers.

²⁸I require at least two observations before treatment and one observation in or after the treatment. For treated firms, coverage is defined relative to the first transfer year. For control firms, I use 2023 as the reference year.

in a panel of 23,367 firm-year observations for 3,729 public and private electric utilities, including 45 treated firms and 3,684 control firms.

I construct two analysis samples from this utility panel. The primary sample contains publicly traded electric utility firms and allows me to examine both financial and electricity based investment outcomes. I restrict the overall utility panel to firms that can be matched to Compustat and have non-missing total assets and sufficient data to construct the control variables. This results in 1,627 firm-year observations for 181 firms, of which 20 are treated. I then construct the cohorts used in the stacked cohort DiD design. Each cohort contains the firms first treated in 2023, 2024 or 2025 and the control firms that do not transfer a credit during the sample period. I restrict each stack to the event window from four years before through two years after treatment and require at least two pre-treatment observations and one observation post-treatment. The resulting public firm sample contains 2,479 firm-cohort-year observations representing 430 firm-cohort pairs.

To broaden the analysis beyond publicly traded firms and increase the number of treated utilities, I construct a second sample that also includes private firms. This increases the number of treated firms from 20 to 45 and brings smaller clean energy developers into the analysis, many of which are privately held and therefore absent from Compustat. The broader sample therefore allows me to examine whether the investment response extends beyond larger publicly traded firms. For this sample, however, I can examine only EIA outcomes because financial statement data are not available for the private firms. I apply the same cohort stacking, event window, and coverage requirements as in the public firm sample. The resulting sample contains 58,344 firm-cohort-year observations representing 10,283 firm-cohort pairs. This broader sample complements the primary public firm sample by allowing me to examine changes in generating assets across a substantially larger and more diverse set of electric utilities, including smaller private developers.

4.3 Descriptive Statistics

Table 2 reports descriptive statistics for the seller side samples prior to cohort stacking. Panel A reports the primary sample of public electric utility firms. Average capital expenditures are \$1.88 billion and equal 6 percent of beginning total assets. Gross property, plant, and equipment averages \$30.08 billion. The average firm operates roughly 33 generating units with a total generating capacity of 3,361 MW. Panel B reports the broader sample of public and private electric utilities. Firms in this sample are substantially smaller, operating an average of approximately 5 generating units with 244 MW of capacity. On average, 69 percent of their electricity generation comes from clean energy

sources. Appendix Table B.2 reports corresponding descriptive statistics for the seller side samples after cohort stacking.

5 Results Seller-Side

5.1 Investment Level Responses

Table 3 reports the effect of tax credit transfers on capital expenditures (Panel A) and PPE (Panel B). Columns (1) through (3) use the baseline control group, while Columns (4) and (5) use the entropy balanced control group. Within the baseline set, the first specification includes only cohort-by-firm and cohort-by-year fixed effects, the second adds one-year-lagged controls, and the third restricts the sample to the 2023 cohort, the only cohort with a full post-treatment window; the entropy-balanced set repeats the lagged-control and 2023-cohort specifications. Across the outcomes and specifications, the coefficient on $Treat \times Post$ is consistently positive.

In my preferred specification with entropy balancing and lagged controls, Column (4), capital expenditures increase by 1.2 percentage points of beginning total assets following a firm’s first tax credit transfer. Relative to the sample mean of 6 percent, this corresponds to an increase of approximately 20 percent. Panel B shows a corresponding expansion in the firm’s stock of productive fixed assets. Gross PPE increases by 6.9 percentage points of beginning total assets, or approximately 7 percent relative to the sample mean, and is statistically significant at the 1 percent level. Together, these estimates indicate that firms increase both current capital spending and their stock of productive assets after entering the transfer market. Restricting the sample to the 2023 cohort in Column (5) yields a similar, if anything larger, PPE estimate of 7.5 percentage points, indicating that the main results are not driven by the shorter post-treatment windows of the later treatment cohorts.

Figure 3 presents the corresponding event-study estimates for the primary sample of public electric utilities, using the baseline and entropy-balanced specifications with one-year-lagged controls. Across both outcomes, the pre-treatment coefficients are close to zero and provide no evidence of differential trends before the first tax credit transfer, supporting the parallel trends assumption. Following treatment, the estimates become positive and increase over the observable post-treatment periods. Overall, the event-study evidence is consistent with the main estimates and provides further support for an increase in investment following tax credit transfers.

Table 4 reports complementary evidence using the number of physical generating

units operated by the firm, in Panel A, and further splits this count into clean (Panel B) and non-clean (Panel C) generators (see Appendix Table A.2 for the classification of generator types as clean or non-clean). Within the baseline set, Columns (1) and (2) use the primary sample of public electric utilities with, respectively, no controls and one-year-lagged controls, Column (3) restricts the public-firm sample to the 2023 cohort with lagged controls, and Column (4) extends the lagged-control specification to the broader sample of public and private electric utilities; the entropy-balanced set (Columns (5)-(7)) repeats the lagged-control, 2023-cohort, and public-and-private specifications. In the preferred public-firm specification, Column (5) of Panel A, the coefficient of 0.124 implies an increase of approximately 13 percent in the number of generators following a tax credit transfer. At a sample mean of approximately 33 generators, this corresponds to roughly four additional generating units. Importantly, the result also extends to the broader sample of public and private electric utilities: the entropy-balanced estimate in Column (7) is 0.048, corresponding to an increase of approximately 5 percent, indicating that the increase in generating assets is not limited to the publicly traded firms covered by Compustat.

Because the objective of clean energy tax credits is to encourage investment in clean technologies, an increase in total generating assets is most relevant if it reflects an expansion in clean rather than non-clean generation. The estimated treatment effect on $Treat \times Post$ is positive and statistically significant for clean generators in Panel B across all specifications. In the preferred entropy-balanced specification with lagged controls, the coefficient is 0.139 in Column (5), corresponding to an increase of approximately 15 percent in clean generating units. The estimate remains positive in the broader public-and-private sample in Column (7), where the coefficient of 0.078 corresponds to approximately 8 percent. In contrast, the estimated effects on non-clean generators in Panel C are close to zero and statistically insignificant in every specification. These results indicate that the increase in generating assets following a tax credit transfer is concentrated in clean generating units. This distinction is important because the increase in overall investment reflects expansion in technologies tax credits and the transferability regulation are intended to promote rather than a general increase in generating assets. Figure 4 presents the corresponding event-study estimates for the primary public-firm sample. The pre-treatment coefficients are close to zero across all three outcomes and provide support for the parallel trends assumption. Following treatment, the estimates increase for overall and clean generators, while there is little corresponding movement in non-clean generators. Taken together, these results reject the null in H1 and provide evidence that tax credit transferability increases clean energy investment.

Appendix Table B.4 and Figure B.1 show that the capital expenditure and PPE re-

sults are robust to replacing the lagged controls with pre-treatment averages interacted with year indicators and to measuring both outcomes in logs. Appendix Table B.5 and Figure B.2 report the corresponding robustness checks for the generator outcomes, comparing the public-firm, 2023-cohort, and public-and-private samples side by side; the estimated effects are similar in sign and magnitude to the main specification across overall, clean, and non-clean generators. These tests indicate that the main findings are not sensitive to the control specification, functional form, sample composition, or the shorter post-treatment windows of the later cohorts.

5.2 Cross-Sectional Tests

I next use cross-sectional tests to examine whether the investment response is consistent with the monetization and financing frictions developed above. I first examine the monetization friction. If transferability increases investment by allowing firms to realize the value of credits they cannot readily use themselves, the response should be more pronounced among firms with limited tax capacity before treatment. I measure limited tax capacity using several pretreatment characteristics, including low profitability, greater net operating losses, and the absence of positive pretax income. Panel A of Table 5 reports the results using the entropy-balanced specification with lagged controls. Across these measures, the investment response is stronger among firms with more limited tax capacity. Low-profitability firms, firms with greater NOLs, and firms reporting pretax losses show larger increases in investment than their counterparts with greater tax capacity. The generator estimates point in the same direction, although the differences across groups are less pronounced. Taken together, these patterns are consistent with limited tax capacity constraining firms' ability to monetize clean energy credits and with transferability increasing investment more where this friction is more binding.²⁹

I next examine the financing friction. If transferability increases investment by providing firms with immediate cash and reducing their reliance on external financing, the investment response should be stronger among firms that face greater financing constraints before treatment. I measure financing constraints using several pretreatment characteristics, including firm size, firm age, leverage, the Hadlock-Pierce index, and operating cash flow. Panel B of Table 5 reports the results. Across these measures, the pattern of invest-

²⁹I also examine prior access to tax-equity financing and rate regulation as additional dimensions that may affect firms' ability to realize the value of tax incentives. The results are more mixed across these splits. Firms without prior tax-equity participation exhibit a larger PPE response, but the generator response is larger among firms with prior tax-equity participation. The effects by regulatory status also vary across outcomes.

ment responses is generally consistent with stronger effects among firms facing greater financing constraints. This evidence is consistent with transferability alleviating financing frictions by increasing internal liquidity and reducing firms' reliance on external capital. Taken together with the monetization tests, the cross-sectional results support the view that both frictions are present in the market and that transferability increases investment more where these frictions are more binding.

Finally, I examine whether the investment response varies with firms' existing exposure to clean energy. Unlike the preceding tests, this analysis does not isolate a friction but examines which firms use transferability to expand investment. Firms with less existing clean generation may have greater scope to shift their generating portfolios toward clean energy, while firms already active in clean technologies may be better positioned to deploy additional investment because of their existing expertise and project pipelines. I measure prior clean energy exposure using the firm's pretreatment share of clean generation and the breadth of clean technologies it operates. Panel C of Table 5 reports the results. Overall, the evidence indicates stronger investment responses among firms with less prior exposure to clean energy. This finding is relevant to the policy objective because it suggests that transferability can expand clean energy investment beyond firms that were already heavily invested in clean technologies and encourage investment among firms with greater scope to transition their generating portfolios.

5.3 Investment Quality Responses

I next examine whether the increase in investment following a tax credit transfer is accompanied by a change in investment quality. Table 6 reports the results. Panel A examines the capacity factor, which captures electricity generation relative to available generating capacity. Panel B examines average electricity generation per generating unit, and Panel C examines ROA. Within the baseline set, Columns (1)-(2) use Sample 1 (public electric utility firm-cohort-year panel) with, respectively, no controls and one-year-lagged controls; Column (3) restricts Sample 1 to the 2023 cohort with lagged controls; and Column (4) repeats the lagged-control specification on Sample 2 (public and private electric utility firm-cohort-year panel) for Panels A and B. The entropy-balanced set (Columns (5)-(7)) repeats the lagged-control, 2023-cohort, and Sample 2 specifications. For capacity factor, the estimated treatment effects are small and statistically insignificant across specifications and samples. Average generation similarly provides no consistent evidence of a change in operating performance. Taken together, the operational measures provide little evidence that the increase in generating assets following a tax credit transfer is accompa-

nied by either an improvement or deterioration in their realized operating performance.

Panel C examines pretax ROA as a supplementary firm-level measure of investment quality. The treatment effect is positive and statistically significant in the baseline specifications, but the effect disappears after entropy balancing. In the preferred entropy-balanced specification, the coefficient is close to zero and statistically insignificant. Taken together, the results provide no consistent evidence that tax credit transfers affect investment quality.

Combined with the investment-level results, these findings suggest that transferability increases the amount of clean energy investment without a change in the average operating performance of a firms' generating assets over the period I observe. I therefore cannot reject the null hypothesis in [H2](#). The results do not provide clear evidence in favor of either of the competing quality channels developed above. I do not find evidence that the additional resources associated with transferability improve operating performance, but neither do I find evidence that the entry of more marginal projects or reduced monitoring associated with tax-equity financing lowers investment quality.

6 Buyer-Side Responses

In the second part of the paper, I examine how firms participate in and use the new market for transferable tax credits. The buyer-side analysis proceeds in three steps. I first characterize who enters the market, compare tax credit buyers with pre-IRA tax-equity investors, and examine which firm characteristics predict a purchase. I then quantify the direct tax benefit of participation by estimating the effect of a tax credit purchase on firms' effective tax rates. Finally, I examine how buyers describe these transactions in their disclosures to explore whether firms use tax credit purchases for broader sustainability objectives in addition to their tax benefits. The participation and disclosure analyses are primarily descriptive and exploratory, while the effective tax rate analysis uses a stacked cohort difference-in-differences design.

6.1 Market Participation

A central policy objective of transferability is to reduce the frictions that previously limited firms' ability to monetize clean energy tax credits. This objective depends not only on developers being able to sell credits but also on firms willing to purchase them. A broader pool of buyers can increase demand for transferable credits, allowing more developers to monetize their credits and obtain funding for clean energy investment. Before

the reform, the only way for an outside investor to obtain tax benefits from a clean energy project it did not develop was through a tax-equity partnership. These arrangements required an ownership interest in the underlying project as well as substantial legal, tax, and transaction expertise, which concentrated participation among large banks and other financial institutions (Garrett and Shive 2025). Transferability instead allows a firm to purchase a credit directly without owning, operating, or closely monitoring the project. These lower entry barriers can broaden and diversify the set of firms willing to finance clean energy. Whether participation actually broadens is not obvious, however, because firms with greater tax expertise, established relationships, and lower due diligence costs may retain an advantage even in the new market.

I examine market participation in two steps. I first compare firms that purchase transferable tax credits with the tax-equity investors that participated in the market before the IRA. I then compare tax credit buyers with a broader set of potential corporate buyers to examine which pre-market characteristics distinguish firms that enter the transfer market from those that do not. For the first analysis, I begin with my hand-collected sample of firms publicly disclosing a tax credit purchase. Between 2023 and 2025, I identify 138 firms that disclose purchasing at least one transferable tax credit. I complement these data with tax-equity transactions from the GlobalData Advanced Deals database from 2015 through 2025 and match both sets of firms by name to Compustat to obtain industry and financial information. Table 7 describes the corresponding sample construction.

I first compare activity in the transferable credit market with the tax-equity market that preceded it. Figure 5 Panel (a) plots the number of newly formed tax-equity partnerships and the number of firms that disclose at least one tax credit purchase in each year from 2015 through 2025. Before the IRA, the number of new tax-equity partnerships is relatively stable, ranging from 5 to 12 per year. Tax credit buyer participation begins at a similar level in 2023, when 14 firms disclose a purchase, but increases rapidly thereafter to 66 firms in 2024 and 126 firms in 2025.³⁰ By 2025, the number of firms participating in the transferable credit market in a single year is more than ten times the number of new tax-equity partnerships typically formed annually before the IRA. This growth indicates a substantial expansion in activity on the buyer side of the clean energy finance market.³¹

³⁰Appendix Table C.2 reports the underlying annual counts. Because firms can purchase credits in multiple years, the annual buyer counts sum to 206 firm-year observations but represent 136 unique buyers over the sample period.

³¹Tax-equity partnerships are typically long term, so the number of newly formed partnerships in a given year understates the stock of active tax-equity relationships. I therefore interpret this comparison as evidence on the scale of new market activity rather than as a direct comparison of the total number of active participants in the two markets. Even with this distinction, the rapid increase in annual tax credit buyer participation points to a substantial expansion relative to the pre-IRA tax-equity market.

I next compare the composition of participants. Figure 5 Panel (b) reports the industry distribution of tax-equity investors and tax credit buyers. Tax-equity participation is concentrated almost entirely in finance and insurance, whereas tax credit buyers span 13 additional industries. Finance and insurance remains the largest individual sector among buyers, but manufacturing and retail trade also account for substantial participation. The buyer pool therefore includes both firms with a traditional presence in energy finance and firms well outside the historical tax-equity investor base. Taken together, the growth in market activity and the expansion across industries are consistent with transferability broadening and diversifying participation in clean energy finance. These comparisons are descriptive and rely on publicly disclosed transactions, so I do not interpret them as causal estimates of the effect of transferability.

I next compare tax credit buyers with corporations that could plausibly have entered the transfer market but did not disclose a purchase. I begin with Compustat firms operating in the same four-digit SIC industries as observed buyers. This restriction makes the comparison firms more similar to actual buyers in terms of their economic environment and industry characteristics. It also allows me to compare firms within industries in which I observe at least one tax credit purchase. I require nonmissing financial data and sufficient panel coverage, defined as at least three pre-market observations between 2018 and 2022 and at least two post-market observations between 2023 and 2025. The resulting sample contains 134 firms that purchase a tax credit between 2023 and 2025 and 517 comparison firms that do not disclose a purchase.³²

Table 9 compares the two groups using characteristics measured before the transfer market opened. I organize the characteristics around tax capacity, tax sophistication, environmental orientation, and general firm attributes. This comparison shows that future buyers exhibit substantially greater tax capacity. Pretax income averages 8 percent of assets among buyers compared with 3 percent among non-buyers, while cash taxes average 2 percent and 1 percent of assets, respectively. I also find that buyers are also more internationally active, with substantially greater foreign operations, whereas prior tax-equity participation and effective tax rates do not differ meaningfully across the two groups. Among the environmental characteristics I find that future buyers have a higher ESG score with an average ESG score of 47.9 compared with 39.9 for non-buyers. Buyers also exhibit higher leverage and ROA, while average firm size is similar across the two groups. Overall, this descriptive evidence suggests that firms entering the transfer market have greater capacity to absorb tax credits and differ along several dimensions that may affect the benefits and costs of participation. I next use these pre-market characteristics to

³²Table 8 Panel A reports summary statistics for the corresponding firm-year panel.

examine which factors predict a firm's decision to purchase a transferable credit.

Table 10 next examines which pre-market characteristics predict a firm's decision to purchase a transferable tax credit. Because firms first enter the market in 2023, 2024, or 2025, I construct separate two-period cohort stacks containing the year immediately before each cohort's purchase year and the purchase year itself. Each stack compares firms first purchasing in that year with firms that do not purchase during the sample period, and I then combine the stacks for estimation. The results point to firms' tax capacity and profitability as predictors of participation. Cash taxes relative to assets are positively associated with the probability of a purchase, particularly when tax capacity and environmental characteristics are considered jointly. ROA is also positive and statistically significant across most specifications. In contrast, the measures of tax sophistication provide mixed evidence. Prior tax-equity participation is negatively associated with entry, while larger book-tax differences are also associated with a lower probability of purchase. Although future buyers have higher ESG scores in the univariate comparisons, ESG scores do not significantly predict participation once other firm characteristics are included. Overall, these results suggest that firms' ability to use the credits they purchase is an important determinant of market entry, while providing less evidence that prior tax sophistication or environmental orientation independently explains participation.

6.2 Effective Tax Rates

I next examine the direct tax benefit firms obtain from purchasing transferable credits. A buyer purchases a credit below face value and applies its full face value against federal income tax liability. The economic benefit to the buyer therefore comes from the discount between the amount paid and the tax liability eliminated. Transferable credits typically trade for approximately 92.5 to 95 cents per dollar of face value (Crux 2025). For example, a firm that pays \$0.925 for a \$1.00 credit reduces its tax liability by \$1.00 while paying \$0.925 to the seller, generating a gross benefit of \$0.075 before transaction costs. This creates a new tax planning opportunity that does not require the buyer to own or operate the underlying clean energy project (Wilde and Wilson 2018).³³

The effect on a firm's effective tax rate depends on the size of the purchase relative to its overall tax position and on how the credit affects the components of tax expense. Because transferred credits offset federal income tax liability, I examine total GAAP ETR, cash ETR, federal GAAP ETR, and the current and deferred components of tax expense.

³³The use of transferred credits is subject to the general business credit limitation and applicable carryback and carryforward provisions.

Total GAAP ETR captures total income tax expense relative to pretax income, while cash ETR captures cash taxes paid relative to pretax income. Federal GAAP ETR isolates federal income tax expense and therefore more closely targets the jurisdiction in which the transferred credit is used. I also separately examine current and deferred ETRs to assess whether any effect is concentrated in current tax expense rather than deferred taxes. How large any change is remains unclear ex ante because the economic gain from the transaction is only the discount on the credit and may be small relative to the buyer's pretax income and overall tax expense.

To estimate the effect of a purchase on effective tax rates, I use the buyer and comparison sample described above and construct a stacked cohort panel around firms' first tax credit purchases. Following the seller-side design in Section 4.1, I separately construct cohorts for firms first purchasing credits in 2023, 2024, and 2025 and compare them with firms that do not purchase a credit during the sample period. I then combine the cohort-specific samples and estimate the change in effective tax rates following the first purchase relative to the change among comparison firms. The regressions include one-year-lagged controls for firm size, ROA, leverage, and net operating losses.³⁴

Tables 11 and ?? reports the results. I find no consistent change in either total GAAP ETR or cash ETR following a firm's first tax credit purchase. In contrast, the more targeted federal ETR measure shows a modest decline following the purchase, with the effect concentrated in the first post-purchase period. This pattern is consistent with transferable credits reducing federal tax expense while having a relatively small effect on the firm's overall effective tax rate. The magnitude is also economically plausible given that the buyer's net benefit is the discount rather than the face value of the credit. For example, purchasing a \$50 million credit at a 10 percent discount generates a \$5 million benefit. For a firm with \$2 billion of pretax income, this corresponds to approximately 0.25 percentage points of pretax income. Overall, the results suggest that transferable credits provide buyers with a measurable but modest tax benefit relative to their broader tax position. I need to work on these tables again. If I do not add the control group and just estimate within treated firms regression, I get negative significant coefficients for Federal ETR and current ETR. So the nulls here might be driven by the control group, I will maybe try some different entropy balancing.

³⁴The specification is

$$Y_{ict} = \beta(Treat_{ic} \times Post_{ct}) + \gamma'X_{ict} + \lambda_{ic} + \omega_{ct} + \epsilon_{ict}, \quad (2)$$

where λ_{ic} and ω_{ct} denote firm-by-cohort and year-by-cohort fixed effects, respectively, and X_{ict} includes one-year-lagged log assets, ROA, leverage, and NOL.

6.3 Disclosures

To examine whether buyers obtain benefits beyond these direct tax savings, I also study how firms describe their tax credit purchases. A tax credit purchase can associate a buyer with clean energy investment and provide content that can be incorporated into broader sustainability narratives (Lin et al. 2024). Because the buyer does not own or operate the underlying project, however, the connection between the purchase and the environmental benefits generated by the project, such as emissions reductions, is indirect. This creates scope for firms to present tax credit purchases as evidence of environmental commitment even when the transaction has a limited connection to the firm's own environmental impact, raising concerns related to greenwashing (Heath et al. 2023; Bilicka et al. 2025; Duchin et al. 2025). My exploratory review of buyers' disclosures suggests that firms use these transactions in different ways. Many firms describe purchases primarily as a means of managing tax liabilities, while others emphasize their connection to clean energy investment or incorporate the purchases into broader sustainability narratives. In some cases, firms also use relationships with clean energy developers to acquire renewable energy certificates alongside the tax credits. This is where I stopped, will be continued.

7 Conclusion

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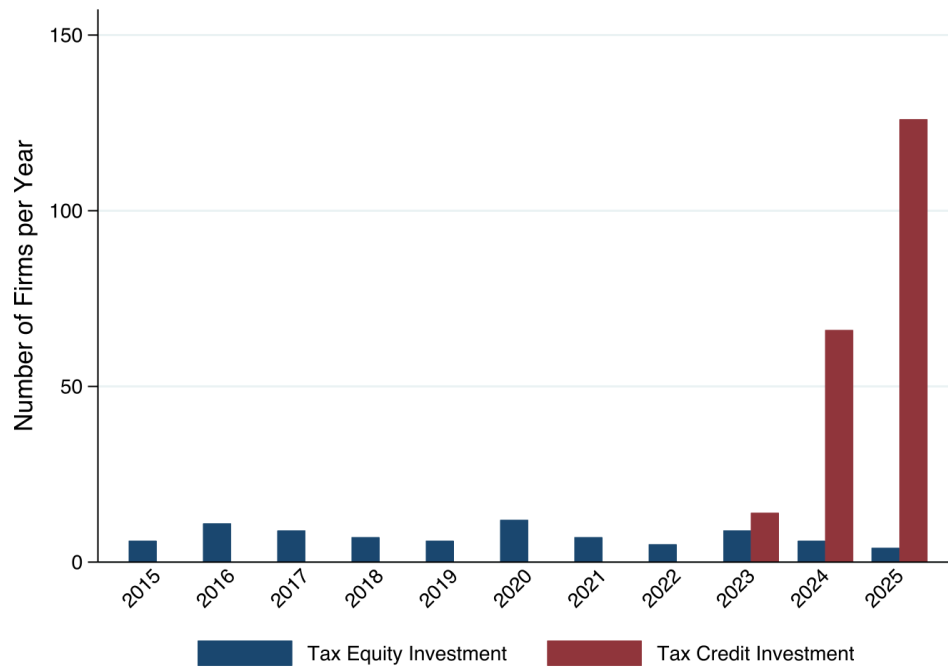
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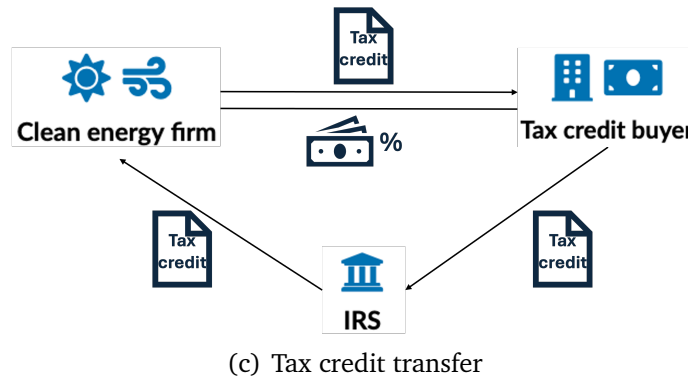
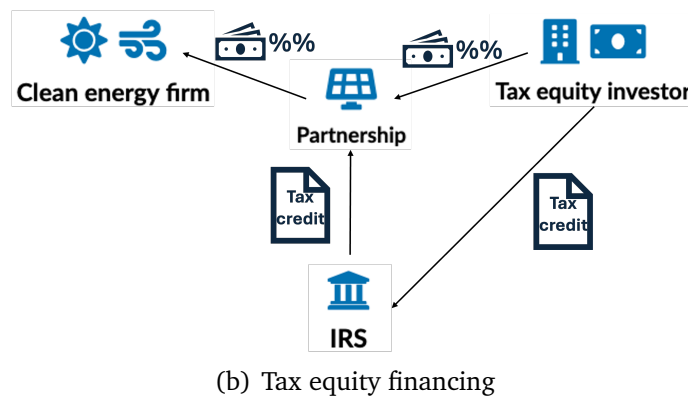
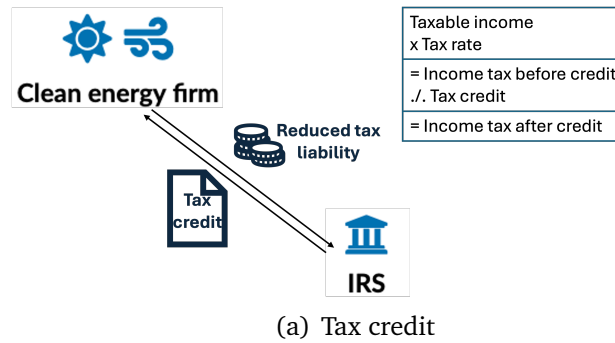
Tables and Figures

Figure 1: Tax Credit Transfer Market



Notes: This figure shows the number of unique firms with tax equity investments (blue) or tax credit investments (red) per year. The sample period is 2015 to 2025. The data is hand-collected from firm disclosures.

Figure 2: Tax Credit Transferability



Notes: This figure illustrates the mechanics of a tax credit transfer under Section 6418 of the Internal Revenue Code, introduced by the Inflation Reduction Act of 2022. Panel (a) depicts a clean energy firm applying a tax credit against its own tax liability. Panel (b) depicts a tax equity financing structure in which a clean energy firm and a tax equity investor form a partnership. Panel (c) depicts a direct transfer, in which a developer lacking sufficient taxable income sells the credit to an unrelated buyer for a discounted cash payment. The buyer claims the full face value against its tax liability, realizing a tax saving equal to the discount. The transaction monetizes the credit without a tax equity partnership and gives the buyer a return without ownership or operational involvement in the underlying project.

Table 1: Seller-Side Responses – Main Sample Construction

	Observations	Firms
<i>Hand-collected tax credit seller sample 2023–2025</i>		
All tax credit seller firm-years	240	80
of which: electric utility firms	150	50
of which: public electric utility firms	95	21
of which: private electric utility firms	55	29
of which: non-utility firms	78	26
<i>Electric utility sample</i>		
Tax credit sellers electric utility firm-years (2023–2025)	150	50
Plus EIA 860/923 electric utility data (2018–2025)	37,757	6,881
Less tax-exempt firms (municipal, co-ops, public power)	(4,403)	(601)
Less firms with insufficient coverage	(6,362)	(2,601)
Less observations with missing data to construct control variables	(1,119)	(1,115)
Public and private electric utility firm-year panel	23,367	3,729
of which: treated firms	288	45
of which: control firms	23,079	3,684
<i>Sample 1: Public electric utility firms</i>		
Public and private electric utility firm-year panel	23,367	3,729
Less private firms	(18,305)	(2,949)
Less observations with missing Compustat match	(1,715)	(526)
Less observations with missing total assets	(1,687)	(564)
Less observations with missing data to construct control variables	(33)	(33)
Public electric utility firm-year panel	1,627	181
of which: treated firms	184	20
of which: control firms	1,443	161
Plus cohort stacking	2,886	322
Less outside event time window [-4, +2]	(1,763)	(0)
Less insufficient per-stack coverage	(271)	(73)
Public electric utility firm-cohort-year panel	2,479	430
<i>Sample 2: Public and private electric utility firms</i>		
Public and private electric utility firm-year panel	23,367	3,729
Plus cohort stacking	46,158	7,368
Less outside event time window [-4, +2]	(8,148)	(4)
Less insufficient per-stack coverage	(3,033)	(810)
Public and private electric utility firm-cohort-year panel	58,344	10,283

Notes: The table reports seller-side sample construction, beginning with the hand-collected tax credit seller sample (2023–2025). For the electric utility samples, I merge U.S. Energy Information Administration data and retain electric utility firms; Sample 1 (public utilities) additionally requires a Compustat match, while Sample 2 (public and private utilities) omits this restriction. Adequate coverage requires at least two pre-years and one post-year relative to the first transfer year (2023 for controls). Each treated cohort forms one cohort stack paired with all never-treated controls over event time [-4, +2].

Table 2: Seller-Side Responses – Summary Statistics for the Firm-Year Panel

	N	Mean	SD	P25	P50	P75
Panel A: Public electric utility firm-year panel						
<i>Dependent variables:</i>						
CapEx	1,627	1,884.76	2,874.72	293.76	811.00	2,220.00
CapEx investment	1,627	0.06	0.04	0.03	0.06	0.08
PPE	1,589	30,076.82	52,148.29	5,738.00	13,231.00	33,000.90
PPE investment	1,589	0.99	0.71	0.71	1.03	1.21
ROA	1,627	0.05	0.08	0.02	0.04	0.06
Generators	1,627	32.82	54.13	3.00	9.00	41.00
Total capacity	1,627	3,361.03	6,710.94	31.90	325.30	3,491.70
Clean generators	1,627	17.77	38.63	0.00	3.00	13.00
Non-clean generators	1,627	15.02	26.63	0.00	4.00	18.00
Clean capacity	1,627	981.28	2,712.54	0.00	32.40	486.60
Non-clean capacity	1,627	2,416.91	5,112.07	0.00	88.00	2,354.60
Capacity factor	1,627	0.40	0.21	0.25	0.39	0.53
Average generation	1,627	324.55	547.24	20.88	135.60	417.59
<i>Control variables:</i>						
Log(Total assets _{<i>t</i>-1})	1,627	9.65	1.53	8.74	9.59	10.66
ROA _{<i>t</i>-1}	1,627	0.05	0.09	0.02	0.04	0.06
Leverage _{<i>t</i>-1}	1,627	0.36	0.15	0.28	0.34	0.42
Log(Net generation _{<i>t</i>-1} + 1)	1,627	13.46	3.55	11.37	13.62	16.26
Pct. clean generation _{<i>t</i>-1}	1,627	0.38	0.43	0.00	0.13	1.00
<i>Cross-sectional variables:</i>						
NOL/assets	593	0.20	0.51	0.01	0.05	0.15
Pretax income	1,627	2,454.15	6,776.24	151.18	513.00	1,727.00
Ever tax equity	1,627	0.04	0.20	0.00	0.00	0.00
Rate regulated	1,597	0.38	0.48	0.00	0.00	1.00
Total assets	1,627	50,115.65	110,093.79	6,646.10	15,222.20	44,876.00
Firm age	1,627	43.59	23.45	21.00	51.00	67.00
HP index	1,627	-4.31	0.43	-4.64	-4.64	-3.96
OCF	1,627	3,700.37	9,743.57	413.00	980.45	3,122.00
Panel B: Public and private electric utility firm-year panel						
<i>Dependent variables:</i>						
Generators	23,367	5.30	17.96	1.00	1.00	3.00
Total capacity	23,367	244.01	796.09	3.00	16.00	107.90
Clean generators	23,367	3.39	14.30	1.00	1.00	2.00
Non-clean generators	23,367	1.91	8.50	0.00	0.00	1.00
Clean capacity	23,367	77.33	285.39	0.60	3.20	30.00
Non-clean capacity	23,367	157.17	592.70	0.00	0.00	7.20
Capacity factor	23,367	0.31	0.22	0.17	0.24	0.43
Average generation	23,367	157.01	325.61	3.41	13.95	146.04
<i>Control variables:</i>						
Net generation _{<i>t</i>-1}	23,367	786,514.65	2,754,134.95	4,590.00	30,836.00	281,167.00
Pct. clean generation _{<i>t</i>-1}	23,367	0.69	0.46	0.00	1.00	1.00

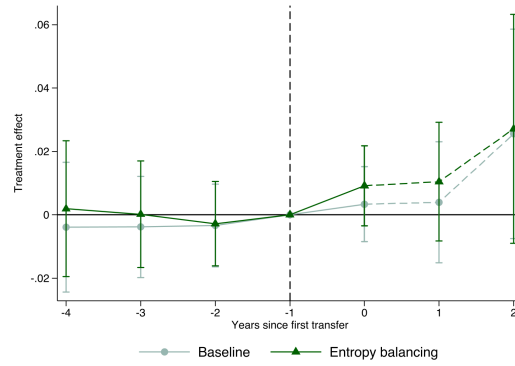
Notes: The table reports summary statistics on the raw firm-year panel, before cohort stacking. Panel A shows summary statistics for Sample 1 (public electric utility firm-year panel) and Panel B for Sample 2 (public and private electric utility firm-year panel). Financial variables are in \$ millions, generation is in GWh and capacity in MW. All variables are defined in Appendix B.1.

Table 3: Seller-Side Responses – Effect of Tax Credit Transfer on Investment Level

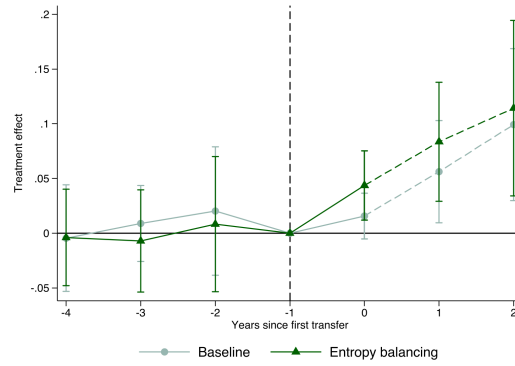
	(1)	(2)	(3)	(4)	(5)
	Baseline		Entropy balanced		
Panel A:	CapEx investment				
Treat × Post	0.008 (1.04)	0.009 (1.15)	0.022*** (2.71)	0.012* (1.71)	0.023** (2.54)
Log(Total assets) _{<i>t</i>−1}		-0.010 (-1.37)	-0.011* (-1.91)	0.010 (0.71)	-0.012 (-0.85)
Pretax income / assets _{<i>t</i>−1}		0.017 (1.04)	0.014 (0.82)	0.045* (1.94)	-0.010 (-0.27)
Leverage _{<i>t</i>−1}		-0.027** (-2.37)	-0.029** (-2.56)	-0.019 (-1.08)	-0.048** (-2.47)
Observations	2,479	2,479	1,087	2,479	1,087
Adj. R ²	0.750	0.754	0.749	0.624	0.677
Panel B:	PPE investment				
Treat × Post	0.029* (1.88)	0.033 (1.65)	0.051*** (2.79)	0.069*** (2.64)	0.075*** (2.91)
Log(Total assets) _{<i>t</i>−1}		-0.381*** (-6.29)	-0.381*** (-4.95)	-0.331*** (-4.61)	-0.430*** (-4.29)
Pretax income / assets _{<i>t</i>−1}		-0.051 (-0.72)	-0.112 (-1.31)	0.052 (0.33)	-0.070 (-0.38)
Leverage _{<i>t</i>−1}		0.026 (0.30)	0.123 (0.79)	0.165 (1.06)	-0.052 (-0.38)
Observations	2,433	2,433	1,068	2,433	1,068
Adj. R ²	0.939	0.953	0.935	0.946	0.953
Sample	Public	Public	2023	Public	2023
Cohort × firm FE	X	X		X	
Cohort × year FE	X	X		X	
Firm FE			X		X
Year FE			X		X
Lagged controls		X	X	X	X

Notes: This table reports stacked-cohort difference-in-differences estimates of the effect of tax credit transfers on investment. The dependent variable is CapEx scaled by one-year-lagged total assets in Panel A and PPE scaled by one-year-lagged total assets in Panel B. Columns (1) through (3) use the baseline control group, whereas columns (4) and (5) use entropy balancing. Within the baseline set, the first column includes only cohort $\times$ firm and cohort $\times$ year fixed effects and uses Sample 1, the public electric utility firm-cohort-year panel; the second adds one-year-lagged control variables, also on Sample 1; the third restricts Sample 1 to the 2023 cohort, the only cohort with a full post-treatment window, with lagged controls and (since a single cohort collapses cohort $\times$ firm and cohort $\times$ year fixed effects to plain firm and year fixed effects) firm and year fixed effects in place of the cohort-interacted terms. The entropy-balanced set repeats the lagged-control specification (column (4)) and the 2023-cohort specification (column (5)); the entropy-balanced, no-controls specification is omitted. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the firm level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 3: Seller-Side Responses – Effect of Tax Credit Transfer on Investment Level, Event Studies



(a) CapEx investment



(b) PPE investment

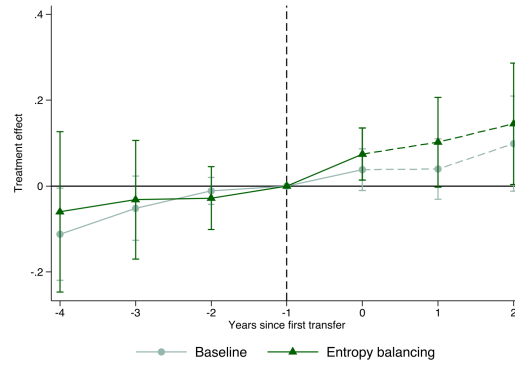
Notes: This figure plots event-study estimates of the effect of tax credit transfers on investment. The estimates are obtained by replacing the *Post* indicator with separate relative-year indicators. The dependent variables are CapEx scaled by one-year-lagged total assets in Panel (a) and PPE scaled by one-year-lagged total assets in Panel (b). Both panels use Sample 1, the public electric utility firm-cohort-year panel. Each regression includes cohort $\times$ firm and cohort $\times$ year fixed effects and one-year-lagged control variables. Each panel presents estimates using the baseline control group (turquoise circles) and entropy balancing (dark green triangles). The series are dashed at event years $t + 1$ and $t + 2$ because the 2024 and 2025 cohorts are only partially represented or not represented at these horizons. Capped vertical lines indicate 95% confidence intervals. Standard errors are clustered at the firm level.

Table 4: Seller-Side Responses – Effect of Tax Credit Transfer on Generator Investment

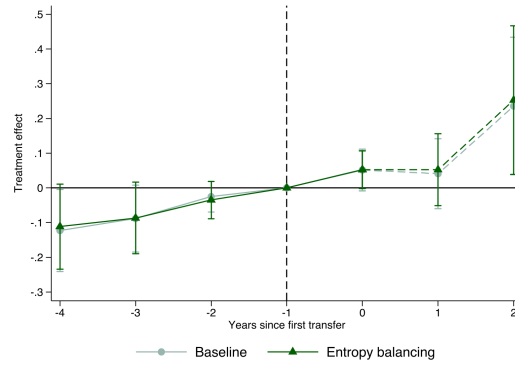
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Baseline				Entropy balanced		
Panel A:	Log(Generators + 1)						
Treat × Post	0.085** (2.47)	0.088** (2.58)	0.097** (2.18)	0.066** (2.51)	0.124*** (2.80)	0.113* (1.91)	0.048* (1.71)
Log(Net gen + 1) _{t-1}		0.022** (2.39)	0.032*** (2.67)	0.014*** (7.23)	0.044 (1.49)	0.105*** (3.34)	0.068*** (3.87)
Pct clean gen _{t-1}		0.177 (1.14)	0.120 (0.68)	-0.061*** (-3.65)	0.144 (0.51)	0.288 (0.74)	-0.292* (-1.94)
Observations	2,479	2,479	1,087	58,344	2,479	1,087	58,344
Adj. R ²	0.984	0.984	0.982	0.983	0.977	0.970	0.990
Panel B:	Log(Clean generators + 1)						
Treat × Post	0.129** (2.17)	0.126** (2.22)	0.165** (2.05)	0.128*** (3.54)	0.139** (2.17)	0.151* (1.71)	0.078* (1.95)
Log(Net gen + 1) _{t-1}		-0.001 (-0.05)	0.003 (0.17)	0.008*** (3.87)	-0.010 (-0.50)	0.035* (1.76)	0.043** (2.21)
Pct clean gen _{t-1}		0.491*** (2.65)	0.450** (2.37)	0.014 (0.52)	0.656*** (2.92)	1.155*** (5.02)	0.002 (0.01)
Observations	2,479	2,479	1,087	58,344	2,479	1,087	58,344
Adj. R ²	0.984	0.984	0.982	0.980	0.987	0.983	0.987
Panel C:	Log(Non-clean generators + 1)						
Treat × Post	-0.008 (-0.17)	-0.002 (-0.05)	0.033 (0.74)	0.000 (0.02)	0.030 (0.51)	0.063 (0.89)	0.005 (0.45)
Log(Net gen + 1) _{t-1}		0.028*** (2.89)	0.038*** (2.95)	0.007*** (4.30)	0.028 (0.72)	0.075 (1.62)	0.010*** (2.68)
Pct clean gen _{t-1}		-0.126 (-0.68)	-0.194 (-0.93)	-0.083*** (-3.43)	0.031 (0.07)	-0.422 (-0.73)	-0.119** (-2.04)
Observations	2,479	2,479	1,087	58,344	2,479	1,087	58,344
Adj. R ²	0.981	0.981	0.979	0.992	0.976	0.969	0.998
Sample	Public	Public	2023	+ Private	Public	2023	+ Private
Cohort × firm FE	X	X		X	X		X
Cohort × year FE	X	X		X	X		X
Firm FE			X			X	
Year FE			X			X	
Lagged controls		X	X	X	X	X	X

Notes: This table reports stacked-cohort difference-in-differences estimates of the effect of tax credit transfers on the number of generators, all as log(count + 1): overall generators (Panel A), and split into clean (Panel B) and non-clean (Panel C) generators. Columns (1) through (4) use the baseline control group, whereas columns (5) through (7) use entropy balancing. Within the baseline set, the first column includes only cohort × firm and cohort × year fixed effects and uses Sample 1, the public electric utility firm-cohort-year panel; the second adds one-year-lagged control variables, also on Sample 1; the third restricts Sample 1 to the 2023 cohort, the only cohort with a full post-treatment window, with lagged controls and (since a single cohort collapses cohort × firm and cohort × year fixed effects to plain firm and year fixed effects) firm and year fixed effects in place of the cohort-interacted terms; the fourth uses the same lagged-control specification as the second column but is estimated on Sample 2, the public and private electric utility firm-cohort-year panel. The entropy-balanced set repeats the lagged-control, 2023-cohort, and Sample 2 specifications (columns (5)-(7)); the entropy-balanced, no-controls specification is omitted. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the utility level are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

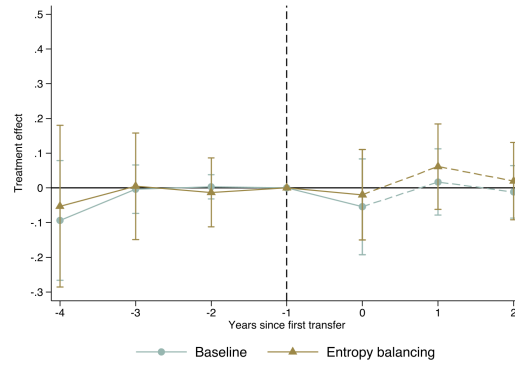
Figure 4: Seller-Side Responses – Effect of Tax Credit Transfer on Generator Investment, Event Studies



(a) Log(Generators + 1)



(b) Log(Clean generators + 1)



(c) Log(Non-clean generators + 1)

Notes: This figure plots event-study estimates of the effect of tax credit transfers on the number of generators: overall (Panel (a)), and split into clean (Panel (b)) and non-clean (Panel (c)) generators, all as $\log(\text{count} + 1)$, obtained by replacing the *Post* indicator with separate relative-year indicators. All panels use Sample 1, the public electric utility firm-cohort-year panel, and include cohort $\times$ firm and cohort $\times$ year fixed effects and one-year-lagged control variables. Each panel presents the baseline control group (turquoise circles) and entropy balancing, shown in dark green for Panels (a)-(b) and dark brown for Panel (c). The series are dashed at event years $t + 1$ and $t + 2$ because the 2024 and 2025 cohorts are only partially represented or not represented at these horizons. Capped vertical lines indicate 95% confidence intervals; standard errors are clustered at the firm level.

Table 5: Seller-Side Responses – Cross-Sectional Tests of Investment Response Channels

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Entropy balanced										
Panel A:	<i>Monetization-Constraint Channel</i>									
	ROA		NOL/assets		Pretax income		Ever tax equity		Rate regulated	
	Low	High	High	Low	Loss	Profit	No	Yes	No	Yes
<i>CapEx investment</i>										
Treat × Post	0.017*	0.003	0.070***	-0.022*	0.031*	0.006	0.009	0.058***	0.003	0.016*
	(1.94)	(0.28)	(5.66)	(-1.95)	(1.95)	(0.80)	(1.37)	(16.69)	(0.32)	(1.69)
p-value, difference	0.660		<0.001***		0.517		0.158		0.068*	
Observations	823	1,656	315	623	200	2,279	2,377	102	1,502	935
Adj. R ²	0.616	0.632	0.911	0.905	0.683	0.605	0.661	0.939	0.731	0.399
<i>PPE investment</i>										
Treat × Post	0.073*	0.039***	0.280***	0.295***	0.157*	0.028**	0.070***	0.042***	0.121**	0.031**
	(1.95)	(3.53)	(5.36)	(2.87)	(2.04)	(2.42)	(2.61)	(6.41)	(2.43)	(2.48)
p-value, difference	0.256		0.332		0.558		0.497		0.236	
Observations	795	1,638	315	594	189	2,244	2,342	91	1,462	935
Adj. R ²	0.946	0.963	0.937	0.935	0.902	0.950	0.946	0.987	0.929	0.930
<i>Log(Generators + 1)</i>										
Treat × Post	0.125**	0.098	0.178**	0.161***	0.140	0.097**	0.124***	0.162**	0.211**	0.068
	(2.17)	(1.62)	(2.41)	(6.16)	(1.58)	(1.99)	(2.67)	(3.00)	(2.34)	(1.49)
p-value, difference	0.125		0.102		0.195		0.944		0.405	
Observations	823	1,656	315	623	200	2,279	2,377	102	1,502	935
Adj. R ²	0.979	0.975	0.994	0.998	0.993	0.966	0.975	0.994	0.967	0.987
Sample	Public	Public	Public	Public	Public	Public	Public	Public	Public	Public
Cohort × firm FE	X	X	X	X	X	X	X	X	X	X
Cohort × year FE	X	X	X	X	X	X	X	X	X	X
Lagged controls	X	X	X	X	X	X	X	X	X	X

Table 5: Seller-Side Responses – Cross-Sectional Tests of Investment Response Channels (continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Entropy balanced										
Panel B:	<i>Financing-Friction Channel</i>									
	Firm size		Firm age		Leverage		HP index		Oper. CF	
	Small	Large	Young	Old	High	Low	High	Low	Neg.	Pos.
<i>CapEx investment</i>										
Treat × Post	0.009 (0.98)	0.016* (1.76)	0.007 (0.68)	0.015 (1.61)	0.025*** (2.80)	-0.001 (-0.10)	0.006 (0.59)	0.015* (1.70)	0.010 (1.18)	0.014* (1.89)
p-value, difference	0.176		0.117		0.172		0.135		0.051*	
Observations	829	1,647	842	1,634	821	1,658	823	1,656	107	2,364
Adj. R ²	0.681	0.630	0.747	0.561	0.717	0.639	0.741	0.562	0.869	0.577
<i>PPE investment</i>										
Treat × Post	0.128* (1.91)	0.040** (2.28)	0.115** (2.51)	0.034** (2.41)	0.128*** (2.79)	0.026 (1.63)	0.120* (1.79)	0.027* (1.79)	0.287*** (15.77)	0.036*** (2.75)
p-value, difference	0.711		0.724		0.063*		0.662		0.334	
Observations	800	1,630	825	1,605	792	1,641	806	1,627	96	2,329
Adj. R ²	0.921	0.941	0.938	0.947	0.942	0.952	0.909	0.929	0.782	0.944
<i>Log(Generators + 1)</i>										
Treat × Post	0.144*** (2.72)	0.098* (1.69)	0.129** (2.30)	0.108* (1.80)	0.073 (1.39)	0.151** (2.40)	0.120** (2.05)	0.114* (1.96)	0.293*** (15.03)	0.084* (1.82)
p-value, difference	0.024**		0.452		0.673		0.452		<0.001***	
Observations	829	1,647	842	1,634	821	1,658	823	1,656	107	2,364
Adj. R ²	0.989	0.964	0.981	0.975	0.985	0.974	0.981	0.975	0.993	0.977
Sample	Public	Public	Public	Public	Public	Public	Public	Public	Public	Public
Cohort × firm FE	X	X	X	X	X	X	X	X	X	X
Cohort × year FE	X	X	X	X	X	X	X	X	X	X
Lagged controls	X	X	X	X	X	X	X	X	X	X

Table 5: Seller-Side Responses – Cross-Sectional Tests of Investment Response Channels (continued)

	(1)	(2)	(3)	(4)
Entropy balanced				
Panel C:	<i>Clean-Investment-Opportunity Channel</i>			
	Pre-clean gen		Pre technologies	
	Low	High	Few	Multiple
<i>CapEx investment</i>				
Treat × Post	0.020**	-0.008	0.018**	0.010
	(2.38)	(-0.92)	(2.05)	(1.15)
p-value, difference	0.045**		0.920	
Observations	1,562	917	1,135	1,344
Adj. R ²	0.669	0.631	0.756	0.541
<i>PPE investment</i>				
Treat × Post	0.081**	0.039*	0.100	0.035**
	(2.51)	(1.95)	(1.46)	(2.61)
p-value, difference	0.768		0.866	
Observations	1,551	882	1,089	1,344
Adj. R ²	0.958	0.934	0.932	0.953
<i>Log(Generators + 1)</i>				
Treat × Post	0.076*	0.148**	0.262**	0.073*
	(1.75)	(2.11)	(2.07)	(1.85)
p-value, difference	0.544		0.219	
Observations	1,562	917	1,135	1,344
Adj. R ²	0.983	0.973	0.961	0.980
Sample	Public	Public	Public	Public
Cohort × firm FE	X	X	X	X
Cohort × year FE	X	X	X	X
Lagged controls	X	X	X	X

Notes: This table reports the Treat × Post coefficient from separate subsample regressions testing three channels for the investment response to tax credit transferability, using Sample 1 (public electric utility firm-cohort-year panel). All columns use entropy balancing and one-year-lagged controls. Panel A measures the monetization-constraint channel using tax-capacity measures (ROA, NOL/assets, Pretax income) and firm characteristics (Ever tax equity, Rate regulated). Panel B measures the financing-friction channel using a financial-constraints index (HP index, the Hadlock and Pierce (2010) size-age index; assets are not inflation-adjusted, as this pipeline has no CPI deflator), firm characteristics (Firm size, Firm age, Leverage), and a cash-flow constraint (Oper. CF). Panel C measures the clean-investment-opportunity channel using each firm's pre-treatment clean generation share (Pre-clean gen, split at 0.5) and the number of distinct clean technologies deployed (Pre technologies, split at the sample median). ROA, NOL/assets, Pretax income, Leverage, Pre-clean gen, and Pre technologies are each firm's pre-treatment mean; Firm size, Firm age, HP index, and Oper. CF are measured in event year $t - 1$; all tercile and median cutoffs are computed once on the baseline (non-entropy-balanced) sample and applied identically to the entropy-balanced sample. Each column omits whichever lagged control mechanically overlaps with that column's own split variable. The row labeled "p-value, difference" reports the p-value on the Treat × Post × group interaction from a pooled regression estimated on the two subsamples together, testing whether the Treat × Post coefficient differs between the two groups; the group indicator itself is absorbed by the cohort × firm fixed effects. All variables are defined in Appendix B.1. Standard errors are clustered at the firm level; t-statistics are reported in parentheses beneath the coefficients. Significance levels are based on two-sided t-tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Seller-Side Responses – Effect of Tax Credit Transfer on Investment Quality

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Baseline				Entropy balanced		
Panel A:	Capacity factor						
Treat × Post	0.007 (0.53)	0.009 (0.80)	0.008 (0.62)	0.004 (0.43)	-0.010 (-0.80)	-0.012 (-0.90)	-0.008 (-0.94)
Log(Net gen + 1) _{<i>t</i>−1}		0.017** (2.16)	0.023*** (2.74)	0.008*** (6.87)	0.013** (2.52)	0.019*** (3.64)	0.011*** (5.95)
Pct clean gen _{<i>t</i>−1}		0.046 (1.15)	0.050 (1.12)	-0.008 (-0.70)	0.025 (0.41)	0.187*** (2.85)	-0.013 (-0.52)
Observations	2,479	2,479	1,087	58,344	2,479	1,087	58,344
Adj. R ²	0.864	0.870	0.872	0.877	0.908	0.910	0.916
Panel B:	Log(Average generation + 1)						
Treat × Post	-0.039 (-0.62)	-0.011 (-0.23)	-0.076 (-1.21)	0.079*** (2.95)	-0.076 (-1.30)	-0.128* (-1.95)	0.005 (0.18)
Log(Net gen + 1) _{<i>t</i>−1}		0.160*** (2.78)	0.219*** (3.21)	0.062*** (7.95)	0.126*** (3.18)	0.151** (2.12)	0.123*** (5.90)
Pct clean gen _{<i>t</i>−1}		0.233 (0.77)	0.256 (0.75)	-0.022 (-0.31)	0.126 (0.28)	1.188** (2.25)	-0.446** (-2.34)
Observations	2,479	2,479	1,087	58,344	2,479	1,087	58,344
Adj. R ²	0.946	0.953	0.950	0.958	0.974	0.965	0.980
Panel C:	ROA						
Treat × Post	0.015** (2.24)	0.020** (2.45)	0.021** (2.02)	-	-0.007 (-0.46)	-0.000 (-0.00)	-
Log(Tot assets) _{<i>t</i>−1}		-0.059*** (-3.45)	-0.053*** (-2.98)	-	-0.008 (-0.25)	0.050 (1.60)	-
Pretax inc/assets _{<i>t</i>−1}		0.212*** (2.90)	0.217*** (3.48)	-	0.188*** (7.49)	0.241*** (5.62)	-
Leverage _{<i>t</i>−1}		0.199*** (3.53)	0.154*** (2.97)	-	-0.025 (-0.52)	0.055 (1.06)	-
Observations	2,479	2,479	1,087		2,479	1,087	
Adj. R ²	0.462	0.521	0.510		0.758	0.780	
Sample	Public	Public	2023	+ Private	Public	2023	+ Private
Cohort × firm FE	X	X		X	X		X
Cohort × year FE	X	X		X	X		X
Firm FE			X			X	
Year FE			X			X	
Lagged controls		X	X	X	X	X	X

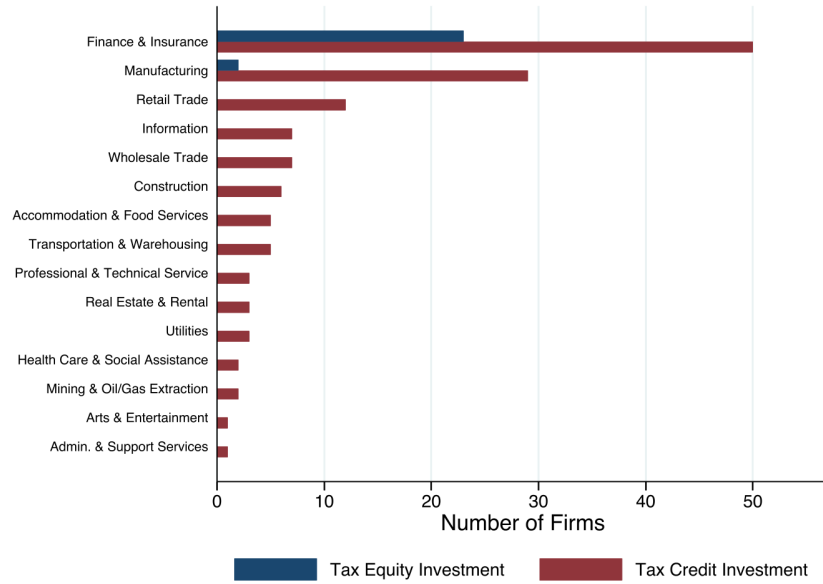
Notes: This table reports stacked-cohort difference-in-differences estimates of the effect of tax credit transfers on investment quality. Columns (1) through (4) use the baseline control group, whereas columns (5) through (7) use entropy balancing. The first column includes only cohort × firm and cohort × year fixed effects and uses Sample 1, the public electric utility firm-cohort-year panel. The second adds one-year-lagged control variables. The third restricts Sample 1 to the 2023 cohort. The fourth is estimated on Sample 2, the public and private electric utility firm-cohort-year panel. The entropy-balanced set repeats the lagged-control, 2023-cohort, and Sample 2 specifications (columns (5)-(7)). Columns (4) and (7) are estimable only in Panels A and B because the Compustat data required to construct ROA are unavailable for private firms in Sample 2. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the firm level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: Buyer-Side Responses – Sample Selection

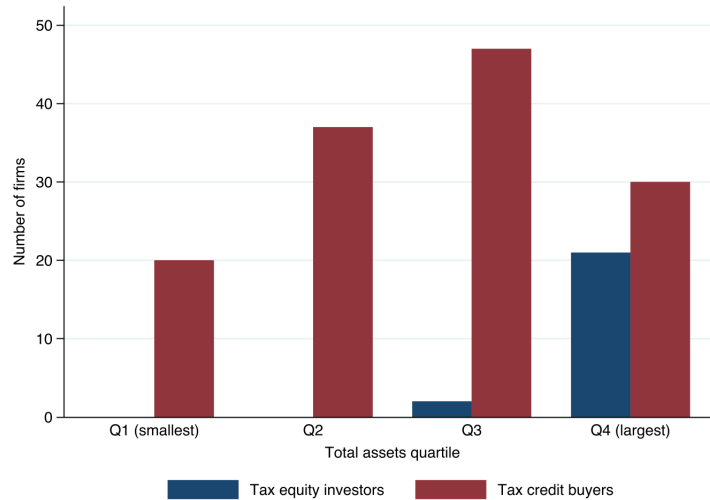
	Observations	Firms
<i>Hand-collected tax credit buyer sample 2023–2025</i>		
All tax credit buyer firm-years	208	138
<i>Sample 1: Tax credit buyer and tax equity firm sample</i>		
Tax credit buyer firm-years (2023–2025)	414	138
Plus tax equity investor indicator (Global Data Advanced Deals, 2015–2025)	396	36
Plus Compustat data (2018–2025)	1,380	174
Less no Compustat SIC match	(112)	(14)
Tax credit buyer and tax equity firm sample	1,268	160
of which: tax credit buyer firms	1,076	136
of which: tax equity investor firms	216	27
including 3 firms that are both tax credit buyers and tax-equity investors		
<i>Sample 2: Tax credit buyer firm-year panel</i>		
Tax credit buyer and tax equity firm sample	1,268	160
Plus control firm-years (same 4-digit-SIC controls, 2018–2025)	7,924	2,377
Less firm-years without Compustat financials	(618)	(74)
Less year 2018 (used for lagged control variables only)	(796)	(37)
Less insufficient coverage	(3,176)	(1,753)
Tax credit buyer firm-year panel	4,428	651
of which: treated firms	931	134
of which: control firms	3,497	517
<i>Sample 3 and 4: Tax credit buyer stacked firm-cohort-year panel</i>		
Tax credit buyer firm-year panel	4,428	651
Plus cohort stacking	6,994	1,034
Less outside event time window [-4, +2]	(1,699)	(0)
Less insufficient per-stack coverage	(338)	(85)
Tax credit buyer stacked firm-cohort-year panel (Sample 4)	9,385	1,600
Tax credit buyer stacked two-period firm-cohort-year panel (Sample 3)	3,160	1,580

Notes: This table reports the sample construction for the hand-collected tax credit buyer sample. Sample 1 (market participation figures) begins with 138 firms disclosing a tax credit purchase between 2023 and 2025, adds tax equity investors observed between 2015 and 2025, and requires a Compustat SIC code match. Sample 2 (cross-sectional determinants and participation tests) adds control firms in the same four-digit SIC industries as observed buyers and requires nonmissing Compustat financials and at least three years between 2019 and 2022 and two years between 2023 and 2025. Samples 3 and 4 stack each treatment cohort (first purchase year 2023, 2024, or 2025) over an event window of four years before through two years after purchase: Sample 3 (determinants analysis) retains only the purchase year and the year before it, while Sample 4 (effective tax rate difference-in-differences analysis) retains the full firm-cohort-year panel.

Figure 5: Buyer-Side Responses



(a) Industry distribution of tax credit and tax equity investment



(b) Firm size

Notes: Panel (a) shows the number of unique firms with tax equity investments (blue) or tax credit purchases (red) per year. Panel (b) shows the number of unique firms per NAICS 2-digit sector for tax equity investors (blue) and tax credit purchasers (red), sorted by total firms (descending); Manufacturing groups NAICS 31–33, Retail Trade groups 44–45, and Transportation & Warehousing groups 48–49. The sample period is 2015 to 2025. The data is hand-collected from firm disclosures.

Table 8: Buyer-Side Responses – Summary Statistics: Firm-Year and Firm-Stack-Year Panels

	N	Mean	SD	P1	P25	P50	P75	P99
Firm-year panel								
Tax equity, ever	4,428	0.04	0.20	0.00	0.00	0.00	0.00	1.00
Total assets	4,428	104,321.70	306,575.38	110.21	2,332.34	7,589.81	36,821.54	1,809,320.00
Firm age	4,428	26.10	17.04	2.00	13.00	23.00	36.00	73.00
Pretax income / assets	4,428	0.04	0.08	-0.10	0.01	0.02	0.04	0.28
ROA	4,428	0.03	0.08	-0.09	0.01	0.01	0.03	0.22
Leverage	4,428	0.13	0.20	0.00	0.02	0.05	0.16	0.83
Cash / assets	4,428	0.11	0.12	0.00	0.03	0.07	0.13	0.63
GAAP ETR	4,355	0.20	0.12	0.00	0.16	0.20	0.24	0.87
Cash ETR	4,337	0.19	0.16	0.00	0.10	0.18	0.24	1.00
Federal GAAP ETR	1,589	0.13	0.13	0.00	0.03	0.12	0.18	0.77
Federal Taxes	1,633	238.34	477.21	-142.25	3.80	50.64	205.95	2,300.81
Current ETR	2,096	0.19	0.16	0.00	0.09	0.19	0.24	1.00
Deferred ETR	2,096	0.07	0.16	0.00	0.00	0.01	0.06	1.00
NOL	4,428	0.03	0.18	0.00	0.00	0.00	0.00	0.62
ESG score	3,035	44.35	18.64	7.32	30.10	41.47	58.61	84.98
Emissions reduction target	3,439	0.15	0.36	0.00	0.00	0.00	0.00	1.00
Firm-stack-year panel								
Tax equity, ever	9,385	0.05	0.21	0.00	0.00	0.00	0.00	1.00
Total assets	9,385	113,021.81	318,424.79	101.90	2,218.23	7,506.81	38,666.86	1,809,320.00
Firm age	9,385	25.70	16.05	2.00	13.00	23.00	35.00	72.00
Pretax income / assets	9,385	0.03	0.09	-0.11	0.01	0.01	0.03	0.27
ROA	9,385	0.02	0.08	-0.10	0.01	0.01	0.02	0.21
Leverage	9,385	0.11	0.17	0.00	0.01	0.04	0.11	0.81
Cash / assets	9,385	0.11	0.12	0.01	0.03	0.07	0.14	0.61
GAAP ETR	9,244	0.20	0.12	0.00	0.16	0.20	0.24	0.87
Cash ETR	9,228	0.19	0.16	0.00	0.10	0.18	0.24	1.00
Federal GAAP ETR	2,780	0.12	0.14	0.00	0.01	0.11	0.18	0.81
Federal Taxes	2,858	239.46	496.05	-116.00	1.30	37.54	195.80	2,300.81
Current ETR	3,953	0.19	0.17	0.00	0.08	0.18	0.24	1.00
Deferred ETR	3,953	0.07	0.16	0.00	0.00	0.01	0.06	1.00
NOL	9,385	0.03	0.19	0.00	0.00	0.00	0.00	0.71
ESG score	6,300	43.11	18.12	7.32	29.73	40.39	55.96	83.81
Emissions reduction target	7,163	0.13	0.33	0.00	0.00	0.00	0.00	1.00

Notes: The table reports N, mean, standard deviation, and percentiles for the firm-year panel and the firm-stack-year panel. Dollar amounts are in \$MM.

Table 9: Buyer-Side Responses – Treated vs. Control: Pre-period Characteristics

	Treated				Control				
	N	Mean	SD	Median	N	Mean	SD	Median	Diff.
Panel A:	<i>Tax capacity</i>								
Pretax income / assets	134	0.08	0.07	0.06	517	0.03	0.06	0.01	0.05***
Cash taxes / assets	132	0.02	0.02	0.01	517	0.01	0.01	0.00	0.01***
Fraction of profitable years	134	0.94	0.14	1.00	517	0.92	0.21	1.00	0.03
Persistent profitability	134	3.74	0.59	4.00	517	3.62	0.85	4.00	0.12
NOL	134	0.03	0.10	0.00	517	0.02	0.16	0.00	0.00
Panel B:	<i>Tax sophistication</i>								
Tax equity	134	0.01	0.10	0.00	517	0.01	0.07	0.00	0.00
GAAP ETR	134	0.20	0.07	0.21	515	0.20	0.08	0.20	-0.00
Federal GAAP ETR	103	0.13	0.07	0.13	134	0.12	0.11	0.11	0.01
Cash ETR	132	0.20	0.11	0.19	515	0.19	0.10	0.19	0.01
Foreign operations	134	0.54	0.48	0.75	517	0.16	0.35	0.00	0.37***
Uncertain tax positions	–	–	–	–	–	–	–	–	–
Panel C:	<i>Environmental orientation</i>								
ESG score	121	48.96	19.67	49.90	325	40.32	16.90	37.01	8.63***
Environmental Pillar score	25	1.04	1.10	0.75	170	0.65	0.84	0.75	0.39**
Emissions reduction target	127	0.25	0.40	0.00	381	0.11	0.29	0.00	0.15***
Panel D:	<i>Firm characteristics</i>								
Log assets	134	9.24	1.68	9.17	517	9.15	2.20	8.77	0.09
Cash / assets	134	0.10	0.12	0.06	517	0.11	0.12	0.08	-0.01
Leverage	134	0.27	0.27	0.24	517	0.09	0.14	0.04	0.18***
ROA	134	0.06	0.05	0.05	517	0.02	0.05	0.01	0.04***

Notes: The table reports pre period (2019–2022) firm characteristics for treated and control firms in the estimation sample, grouped into tax capacity, tax sophistication, environmental orientation, and firm characteristics. Diff. is the treated-minus-control difference from a two-sample *t*-test. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 10: Buyer-Side Responses – Determinants of Tax Credit Purchase

	(1)	(2)	(3)	(4)	(5)	(6)
	Firm Char.	Tax Capacity	Tax Soph.	Environmental	Capacity + Env.	Soph. + Env.
<i>Tax credit purchase</i>						
Log(Total assets _{<i>t</i>-1})	0.069*** (2.62)	0.076*** (2.86)	0.132*** (2.69)	0.027 (0.58)	0.038 (0.81)	0.131* (1.93)
ROA _{<i>t</i>-1}	8.888*** (5.96)	6.077*** (3.30)	7.308*** (4.63)	7.732*** (4.48)	5.588** (2.41)	6.589*** (3.75)
Leverage _{<i>t</i>-1}	1.306*** (3.62)	1.170*** (3.21)	0.406 (1.00)	1.072*** (2.66)	0.994** (2.43)	0.323 (0.72)
Cash / assets _{<i>t</i>-1}	-1.165 (-1.58)	-1.392* (-1.75)	-2.204** (-2.10)	-0.813 (-0.96)	-1.010 (-1.15)	-2.195* (-1.86)
NOL _{<i>t</i>-1}		-0.020 (-0.08)			-0.050 (-0.20)	
Cash taxes / assets _{<i>t</i>-1}		12.893* (1.81)			9.483 (1.14)	
Federal GAAP ETR _{<i>t</i>-1}			0.927* (1.71)			1.001 (1.56)
Book-tax difference _{<i>t</i>-1}			-4.447** (-2.52)			-5.288** (-2.35)
Tax equity			-0.976* (-1.79)			-1.157** (-2.04)
ESG score _{<i>t</i>-1}				0.007 (1.38)	0.006 (1.25)	0.002 (0.27)
Emissions reduction target _{<i>t</i>-1}				0.363* (1.87)	0.350* (1.81)	0.357 (1.48)
Observations	1,580	1,580	458	1,085	1,085	372
Pseudo R2	0.213	0.220	0.198	0.222	0.226	0.208
Cohort × period FE	X	X	X	X	X	X
Lagged controls	X	X	X	X	X	X

Notes: The table reports probit estimates of the probability of a tax credit purchase, using a stacked cohort panel restricted to the year before and the year of each cohort's purchase, with control firms replicated once per cohort stack. Column (1) includes baseline firm characteristics (log assets, ROA, leverage, cash/assets); Columns (2)–(4) add tax capacity, tax sophistication, and environmental variables in turn; Columns (5) and (6) report combinations of these groups. Fixed effects are cohort × period. *t*-statistics clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Buyer-Side Responses – DiD Effective Tax Rate After Purchase

	(1)	(2)	(3)	(4)	(5)
	GAAP ETR	Cash ETR	Federal ETR	Current ETR	Deferred ETR
Panel A:					
	<i>Baseline</i>				
Treat × Post	0.005 (0.57)	0.035*** (3.10)	-0.007 (-0.58)	-0.006 (-0.50)	0.017 (1.21)
Log(Total assets _{<i>t</i>-1})	0.026** (2.07)	0.021 (1.40)	0.011 (0.99)	0.012 (0.97)	0.034* (1.75)
ROA _{<i>t</i>-1}	0.010 (0.21)	0.295** (2.39)	0.143 (1.28)	0.108 (1.32)	-0.049 (-0.89)
Leverage _{<i>t</i>-1}	0.183** (2.32)	-0.022 (-0.28)	0.042 (1.21)	0.053 (1.14)	0.215** (2.25)
NOL _{<i>t</i>-1}	0.037 (0.62)	0.034 (0.95)	-0.009 (-0.42)	-0.002 (-0.08)	0.040 (0.68)
Observations	9,196	9,180	2,730	3,934	3,934
Adj. R2	0.229	0.227	0.345	0.280	0.221
Panel B:					
	<i>Entropy Balanced</i>				
Treat × Post	-0.000 (-0.02)	0.033*** (2.67)	-0.011 (-0.85)	-0.007 (-0.47)	-0.006 (-0.45)
Log(Total assets _{<i>t</i>-1})	0.025** (2.06)	0.040** (2.44)	0.020** (2.13)	0.016 (1.39)	0.034 (1.61)
ROA _{<i>t</i>-1}	0.040 (0.78)	0.116 (0.79)	0.061 (0.75)	0.069 (0.76)	-0.044 (-1.41)
Leverage _{<i>t</i>-1}	0.119*** (5.33)	0.049 (0.85)	0.045** (1.98)	0.059** (2.10)	0.097* (1.83)
NOL _{<i>t</i>-1}	0.017 (0.49)	0.131*** (3.75)	0.040*** (4.01)	0.052* (1.65)	-0.028 (-1.10)
Observations	9,196	9,180	2,730	3,934	3,934
Adj. R2	0.257	0.293	0.375	0.368	0.195
Stack x firm FE	X	X	X	X	X
Stack x year FE	X	X	X	X	X
Lagged controls	X	X	X	X	X

Notes: The table reports stacked cohort difference-in-differences estimates of *Treat × Post* on effective tax rates, baseline (unweighted) control group, with the full lagged-control specification (assets, ROA, leverage, NOL, pretax income/assets, all *t* − 1) and stack × firm and stack × year fixed effects. *t*-statistics clustered by firm are in parentheses. *** *p*<0.01, ** *p*<0.05, * *p*<0.1.

Table 12: Buyer-Side Responses – Sustainability Reports

Firm	Year	Excerpt from sustainability report
Tractor Supply	2024	<i>We also support U.S. manufacturing and lower-emission transportation fuels through the purchase of transferable federal tax credits.</i>
Tradeweb Markets Inc.	2024	<i>In 2024, Tradeweb did purchase Transferable Tax Credits from a renewable energy facility. Transferable tax credits may be considered an indirect infrastructure investment as their intention is to allow indirect support for the development of renewable energy projects.</i>
Market Axess Holdings Inc.	2024	<i>In 2024, the Company continued purchasing renewable energy-related transferable tax credits under the Inflation Reduction Act (the "IRA"). The transferability market created by the IRA allows producers of renewable energy or related manufacturing components to sell the tax credits generated by their activities to corporate purchasers. Through the purchase of these credits, the Company has reduced its federal tax liability while supporting innovative companies as they continue to drive the United States' transition to renewable energy.</i>
Nasdaq Inc.	2024	<i>In 2024, Nasdaq purchased transferable tax credits from two producers of renewable energy in the U.S. The transferability market allows producers of renewable energy or related manufacturing components to sell the tax credits generated by their activities to corporate purchasers. Through the purchase of these credits, Nasdaq has reduced its federal tax liability while In 2024, the Company continued purchasing renewable energy-related transferable tax credits under the Inflation Reduction Act (the "IRA"). The transferability market created by the IRA allows producers of renewable energy or related manufacturing components to sell the tax credits generated by their activities to corporate purchasers. Through the purchase of these credits, the Company has reduced its federal tax liability while supporting innovative companies as they continue to drive the United States' transition to renewable energy.</i>
Stifel Financial Corp.	2025	<i>Stifel directly purchased over \$100 million in renewable transfer tax credits to further bolster our own commitment to the energy transition.</i>

Notes: The table shows excerpts from corporate sustainability in which firms disclose the purchase of transferable tax credits.

Appendix

The next pages include the following appendices:

- **Tax Credit Transfer Market** (Appendix [A](#)):

- **Seller-Side Responses** (Appendix [B](#)):

Appendix [B.1](#): Variable Definitions

Appendix [B.2](#): Additional Tables and Figures

- **Buyer-Side Responses** (Appendix [C](#))

Appendix [C.1](#): Variable Definitions

Appendix [C.2](#): Additional Tables and Figures

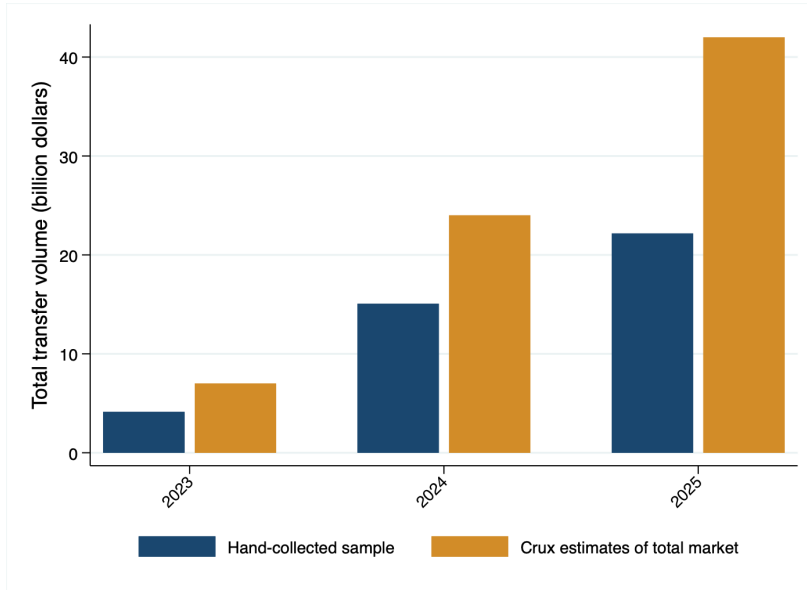
A The New Market for Tax Credits

Table A.1: Transferable Clean Energy Tax Credits

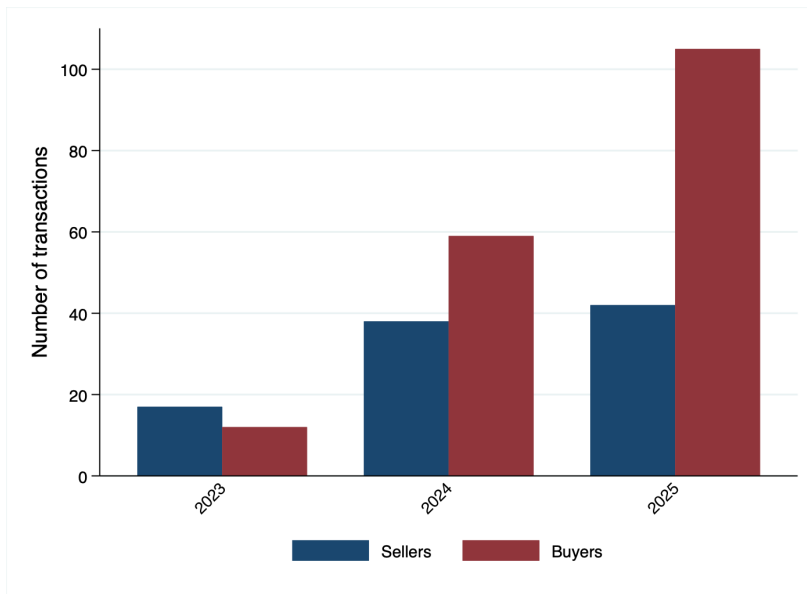
Tax Credit	Code	Technology	Credit Base	Created	First Effective
<i>Production Tax Credit</i>	§45	Wind	Electricity produced	1992	1992
<i>Investment Tax Credit</i>	§48	Solar, storage	Investment costs	1992	1992
<i>Alternative Fuel Refueling Property Credit</i>	§30C	EV charging, alt-fuel refueling	Investment costs	2005	2006
<i>Carbon Capture and Sequestration Credit</i>	§45Q	Carbon capture	CO ₂ captured	2008	2008
<i>Advanced Energy Project Credit</i>	§48C	Clean energy production facilities	Investment costs	2009	2009
<i>Technology-Neutral Clean Electricity Credits</i>	§45Y/§48E	Successors to §45/§48	Investment & production	2022 (IRA)	2025
<i>Clean Hydrogen Production Credit</i>	§45V	Hydrogen	Clean hydrogen produced	2022 (IRA)	2023
<i>Advanced Manufacturing Production Credit</i>	§45X	Clean energy components	Produced goods/sales	2022 (IRA)	2023
<i>Zero-Emission Nuclear Power Credit</i>	§45U	Nuclear plants	Electricity produced	2022 (IRA)	2024
<i>Clean Fuel Production Credit</i>	§45Z	Sustainable fuels	Fuel produced	2022 (IRA)	2025

Notes: This table lists the clean energy tax credits eligible for transfer under Section 6418 of the Internal Revenue Code, introduced by the Inflation Reduction Act (IRA) of 2022, reporting for each its Code section (*Code*), the incentivized technology or activity (*Technology*), the basis on which the credit is computed (*Credit Base*), and the years it was created and became effective.

Figure A.1: Tax Credit Transfers



(a) Benchmarking hand-collected sample to Crux market estimates



(b) Number of tax credit transfers

Notes: Panel (a) compares the total transfer volume (\$ billions) per year in the hand-collected sample to Crux's estimates of the total transferability market, based on its market intelligence reports (Crux 2025); missing transfer prices are imputed using the within credit-type by technology group mean. Panel (b) shows the number of tax credit transfer transactions per year, based on hand-collected transfers from firm disclosures and announcements, separately for sellers and buyers, with each transaction counted once per deal agreement. Panel (b) shows the total transfer value (\$ billions) per year, separately for sellers and buyers. The sample period is 2023 to 2025.

Table A.2: Clean versus Non-clean Energy Generation

Category	Classification	EIA Source Codes
Solar	Clean	SUN
Wind	Clean	WND
Water (hydroelectric)	Clean	WAT, H2
Geothermal	Clean	GEO
Biomass	Clean	LFG, OBG, AB, MSW, OBS, OBL, BLQ, WDL, SLW, WH, WDS
Battery storage	Clean	MWH
Nuclear	Clean	NUC
Coal	Non-clean	RC, LIG, PT, WC, SUB, BIT, SGC
Gas	Non-clean	NG, OG, BFG, PG
Petroleum	Non-clean	SGP, JF, KER, DFO, RFO, W0, PC

Notes: This table reports how generators are classified as clean or non-clean throughout the paper, based on each generator's primary energy source in EIA Form 860 (*EnergySource1*): clean if solar, wind, hydroelectric, geothermal, biomass, battery storage, or nuclear, and non-clean if coal, gas, or petroleum.

B Seller-Side Responses

B.1 Variable Definition Seller-Side Responses

Table B.1: Seller-Side Responses – Variable Definitions

Variable	Definition	Data source
CapEx _{<i>i,t</i>}	Capital expenditures for firm <i>i</i> in year <i>t</i> , in \$ millions.	Compustat (capx)
CapEx investment _{<i>i,t</i>}	CapEx divided by firm <i>i</i> 's total assets at the end of year <i>t</i> – 1.	Compustat (capx/at _{<i>t-1</i>})
PPE _{<i>i,t</i>}	Gross property, plant, and equipment for firm <i>i</i> in year <i>t</i> , in \$ millions.	Compustat (ppegt)
PPE investment _{<i>i,t</i>}	PPE divided by firm <i>i</i> 's total assets at the end of year <i>t</i> – 1.	Compustat (ppegt/at _{<i>t-1</i>})
ROA _{<i>i,t</i>}	Pretax income divided by firm <i>i</i> 's total assets at the end of year <i>t</i> – 1.	Compustat (pi/at _{<i>t-1</i>})
Total assets _{<i>i,t</i>}	Total assets for firm <i>i</i> in year <i>t</i> , in \$ millions.	Compustat (at)
Pretax income / assets _{<i>i,t</i>}	Pretax income divided by total assets for firm <i>i</i> in year <i>t</i> .	Compustat (pi/at)
Leverage _{<i>i,t</i>}	Long-term debt (plus the current portion of long-term debt, where available) divided by total assets for firm <i>i</i> in year <i>t</i> .	Compustat (dltt[+dlc]/at)
Number of generators _{<i>i,t</i>}	Number of distinct electricity generating units operated by firm <i>i</i> in year <i>t</i> .	EIA Form 860
Clean / Non-clean generators _{<i>i,t</i>}	Number of generators using clean (solar, hydroelectric, wind) / non-clean generating technologies operated by firm <i>i</i> in year <i>t</i> .	EIA Form 860
Total capacity _{<i>i,t</i>}	Total nameplate capacity across all generators operated by firm <i>i</i> in year <i>t</i> , in megawatts (MW).	EIA Form 860
Clean / Non-clean capacity _{<i>i,t</i>}	Total nameplate capacity, in MW, across clean / non-clean generators operated by firm <i>i</i> in year <i>t</i> .	EIA Form 860
Net generation _{<i>i,t</i>}	Total net electricity generation across all generators operated by firm <i>i</i> in year <i>t</i> , in gigawatt-hours (GWh).	EIA Form 923
Pct. generation clean _{<i>i,t</i>}	Share of net generation from clean generating technologies (solar, hydroelectric, wind) out of total net generation for firm <i>i</i> in year <i>t</i> .	EIA Forms 860, 923
Capacity factor _{<i>i,t</i>}	Net generation divided by nameplate capacity × 8,760 hours for firm <i>i</i> in year <i>t</i> .	EIA Forms 860, 923
Average generation _{<i>i,t</i>}	Net generation divided by the number of generators, excluding battery storage units, operated by firm <i>i</i> in year <i>t</i> , in gigawatt-hours (GWh).	EIA Forms 860, 923

B.2 Additional Tables and Figures Seller-Side Responses

Table B.2: Seller-Side Responses – Summary Statistics for the Firm-Cohort-Year Panel

	N	Mean	SD	P25	P50	P75
Panel A: Public electric utility firm-cohort-year panel						
<i>Dependent variables:</i>						
CapEx	2,479	2,041.80	2,975.63	329.48	911.00	2,427.00
CapEx investment	2,479	0.06	0.04	0.03	0.06	0.08
Log(CapEx + 1)	2,479	6.61	1.76	5.80	6.82	7.79
PPE	2,433	33,301.87	55,382.22	6,843.00	14,145.70	37,165.00
PPE investment	2,433	0.95	0.41	0.73	1.05	1.21
Log(PPE + 1)	2,433	9.53	1.45	8.83	9.56	10.52
ROA	2,479	0.05	0.07	0.02	0.04	0.06
Generators	2,479	32.41	53.20	3.00	8.00	40.00
Log(Generators + 1)	2,479	2.53	1.39	1.39	2.20	3.71
Total capacity	2,479	3,350.20	6,818.18	32.60	287.60	3,376.70
Log(Total capacity + 1)	2,479	5.71	2.71	3.51	5.67	8.12
Capacity factor	2,479	0.40	0.21	0.24	0.38	0.53
Average generation	2,479	322.10	547.72	19.73	132.74	416.55
Log(Average generation + 1)	2,479	4.47	1.97	3.03	4.90	6.03
<i>Control variables:</i>						
Log(Total assets _{<i>t-1</i>})	2,479	9.77	1.52	8.86	9.69	10.83
ROA _{<i>t-1</i>}	2,479	0.05	0.08	0.02	0.04	0.07
Leverage _{<i>t-1</i>}	2,479	0.36	0.14	0.29	0.35	0.42
Log(Net generation _{<i>t-1</i>} + 1)	2,479	13.43	3.60	11.38	13.59	16.26
Pct. clean generation _{<i>t-1</i>}	2,479	0.37	0.43	0.00	0.10	1.00
<i>Cross-sectional variables:</i>						
Clean generators	2,479	17.64	37.68	0.00	3.00	13.00
Log(Clean generators + 1)	2,479	1.64	1.53	0.00	1.39	2.64
Non-clean generators	2,479	14.76	25.50	0.00	3.00	17.00
Log(Non-clean generators + 1)	2,479	1.74	1.43	0.00	1.39	2.89
Clean capacity	2,479	1,020.81	2,777.16	0.00	31.90	481.20
Log(Clean capacity + 1)	2,479	3.56	3.08	0.00	3.49	6.18
Non-clean capacity	2,479	2,342.77	4,823.16	0.00	100.00	2,487.50
Log(Non-clean capacity + 1)	2,479	4.51	3.46	0.00	4.62	7.82
NOL/assets	938	0.17	0.35	0.01	0.05	0.15
Pretax income	2,479	2,922.42	6,913.29	218.32	627.77	2,211.50
Ever tax equity	2,479	0.04	0.20	0.00	0.00	0.00
Rate regulated	2,437	0.38	0.49	0.00	0.00	1.00
Total assets	2,476	53,327.23	105,041.00	7,753.00	17,221.00	53,951.00
Firm age	2,476	46.36	23.27	23.00	56.00	71.00
HP index	2,479	-4.36	0.39	-4.64	-4.64	-4.08
OCF	2,476	4,830.61	12,895.60	485.00	1,106.90	3,920.00

Table B.2: Seller-Side Responses – Summary Statistics for the Firm-Cohort-Year Panel (continued)

	N	Mean	SD	P25	P50	P75
Panel B: Public and private electric utility firm-cohort-year panel						
<i>Dependent variables:</i>						
Generators	58,344	5.14	17.73	1.00	1.00	3.00
Log(Generators + 1)	58,344	1.20	0.79	0.69	0.69	1.39
Total capacity	58,344	231.41	754.59	3.00	16.00	106.50
Log(Total capacity + 1)	58,344	3.23	2.04	1.39	2.83	4.68
Capacity factor	58,344	0.31	0.22	0.17	0.24	0.43
Average generation	58,344	157.33	326.44	3.38	13.80	144.67
Log(Average generation + 1)	58,344	3.17	2.07	1.48	2.69	4.98
<i>Control variables:</i>						
Log(Net generation _{<i>t</i>-1} + 1)	58,344	10.31	3.21	8.43	10.34	12.54
Pct. clean generation _{<i>t</i>-1}	58,344	0.69	0.46	0.00	1.00	1.00
<i>Cross-sectional variables:</i>						
Clean generators	58,344	3.29	14.15	1.00	1.00	2.00
Log(Clean generators + 1)	58,344	0.81	0.80	0.69	0.69	1.10
Non-clean generators	58,344	1.84	8.33	0.00	0.00	1.00
Log(Non-clean generators + 1)	58,344	0.45	0.81	0.00	0.00	0.69
Clean capacity	58,344	73.41	273.89	0.30	3.10	29.95
Log(Clean capacity + 1)	58,344	2.04	1.94	0.26	1.41	3.43
Non-clean capacity	58,344	148.79	562.82	0.00	0.00	7.20
Log(Non-clean capacity + 1)	58,344	1.36	2.37	0.00	0.00	2.10

Notes: This table reports summary statistics for the stacked-cohort panel, in which each treated cohort is paired with all never-treated control firms and control firm-years are repeated once per cohort stack. Panel A reports Sample 1 (public electric utility firm-cohort-year panel) and Panel B reports Sample 2 (public and private electric utility firm-cohort-year panel). Variables are reported in the form used in the analysis, including level, log, and capacity outcomes, the one-year-lagged form of each control, and the clean/non-clean breakdown of generators and total capacity. Panel A's cross-sectional variables are reported in the same form used to construct the splits in Table 5: ROA, NOL/assets, and pretax income are each firm's pre-treatment mean; total assets, firm age, the HP index, and OCF are measured in event year $t - 1$; ever tax equity and rate regulated are time-invariant indicators. Generation is in GWh, capacity in MW, and financial variables in millions of dollars. All variables are defined in Appendix B.1.

Table B.3: Seller-Side Responses – pre-treatment Comparison of Treatment and Control Firms

	Treatment		Baseline control		Entr. balanced control		Diff. in Means	
	N	Mean	N	Mean	N	Mean	Unweighted	Weighted
Panel A: Public electric utility firm-cohort-year panel (pre-treatment)								
Log(Total assets)	20	9.62	410	9.78	410	9.62	-0.16	-0.00
Pretax income/assets	20	-0.00	410	0.05	410	-0.00	-0.05***	-0.00
Leverage	20	0.35	410	0.32	410	0.35	0.02	0.00
Log(total capacity)	20	7.64	410	5.57	410	7.64	2.06***	0.00
Panel B: Public and private electric utility firm-cohort-year panel (pre-treatment)								
Log(Net generation)	44	13.80	10,239	10.46	10,239	13.79	3.34***	0.01
Pct. clean generation	44	0.58	10,239	0.70	10,239	0.58	-0.12*	-0.00
Log(total capacity)	44	6.34	10,239	3.19	10,239	6.34	3.16***	0.01

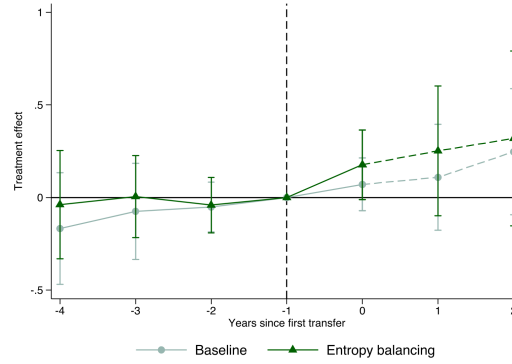
Notes: This table compares treated and control firms before treatment on the entropy balancing target variables for each sample: Panel A reports Sample 1 (public electric utility firm-cohort-year panel) and Panel B reports Sample 2 (public and private electric utility firm-cohort-year panel). Each row is a firm-stack's own pre-treatment mean; the treatment group is unweighted throughout. *Diff. in Means, Unweighted* reports the treated-minus-control difference from a two-sample t-test against the baseline control group; *Weighted* reports the treatment coefficient from a regression against the entropy-balancing-weighted control group, with robust standard errors. All variables are defined in Appendix B.1. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.4: Seller-Side Responses – Robustness for Effect of Tax Credit Transfer on Investment

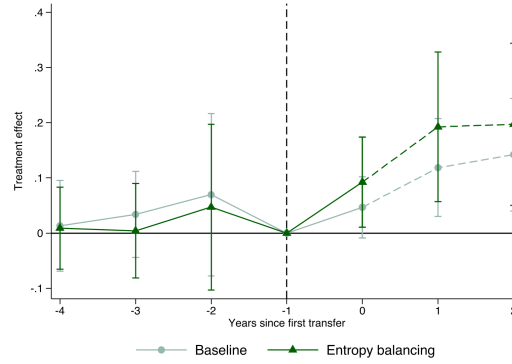
	(1)	(2)	(3)	(4)
Entropy balanced				
Panel A:	CapEx investment		Log(CapEx + 1)	
Treat × Post	0.011*	0.021**	0.242*	0.332***
	(1.78)	(2.51)	(1.94)	(2.64)
Log(Total assets) _{t-1}			1.423***	0.740***
			(4.64)	(4.58)
Pretax income / assets _{t-1}			0.302	0.447
			(0.76)	(1.39)
Leverage _{t-1}			-0.943	-0.649**
			(-1.34)	(-2.33)
Observations	2,479	1,087	2,479	1,087
Adj. R ²	0.682	0.702	0.967	0.972
Panel B:	PPE investment		Log(PPE + 1)	
Treat × Post	0.071***	0.054**	0.127**	0.068**
	(3.30)	(2.38)	(2.47)	(2.51)
Log(Total assets) _{t-1}			0.407***	0.402***
			(2.90)	(3.83)
Pretax income / assets _{t-1}			-0.153	-0.180**
			(-0.59)	(-2.09)
Leverage _{t-1}			0.497*	0.209
			(1.96)	(1.43)
Observations	2,433	1,068	2,433	1,068
Adj. R ²	0.944	0.942	0.995	0.994
Sample	Public	2023	Public	2023
Cohort × firm FE	X		X	
Cohort × year FE	X		X	
Firm FE		X		X
Year FE		X		X
Lagged controls			X	X
Pre-treatment controls	X	X		

Notes: This table reports robustness checks for the main results on capital expenditures (Panel A) and PPE (Panel B). All specifications here use entropy balancing. Columns (1)-(2) report the investment-ratio (level) outcome under the pre-treatment mean-x-year control specification: column (1) on Sample 1 (public electric utility firm-cohort-year panel) and column (2) restricted to the 2023 cohort, the only cohort with a full post-treatment window. Columns (3)-(4) instead use the alternative log(1 + y) transformation of each outcome with one-year-lagged controls: column (3) on Sample 1; column (4) restricted to the 2023 cohort. Both 2023-cohort columns (since a single cohort collapses cohort × firm and cohort × year fixed effects to plain firm and year fixed effects) use firm and year fixed effects in place of the cohort-interacted terms. The log(1 + y), no-controls specification is omitted. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the firm level are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure B.1: Seller-Side Responses – Effect of Tax Credit Transfer on Investment using Logs, Event Studies



(a) $\text{Log}(\text{CapEx} + 1)$



(b) $\text{Log}(\text{PPE} + 1)$

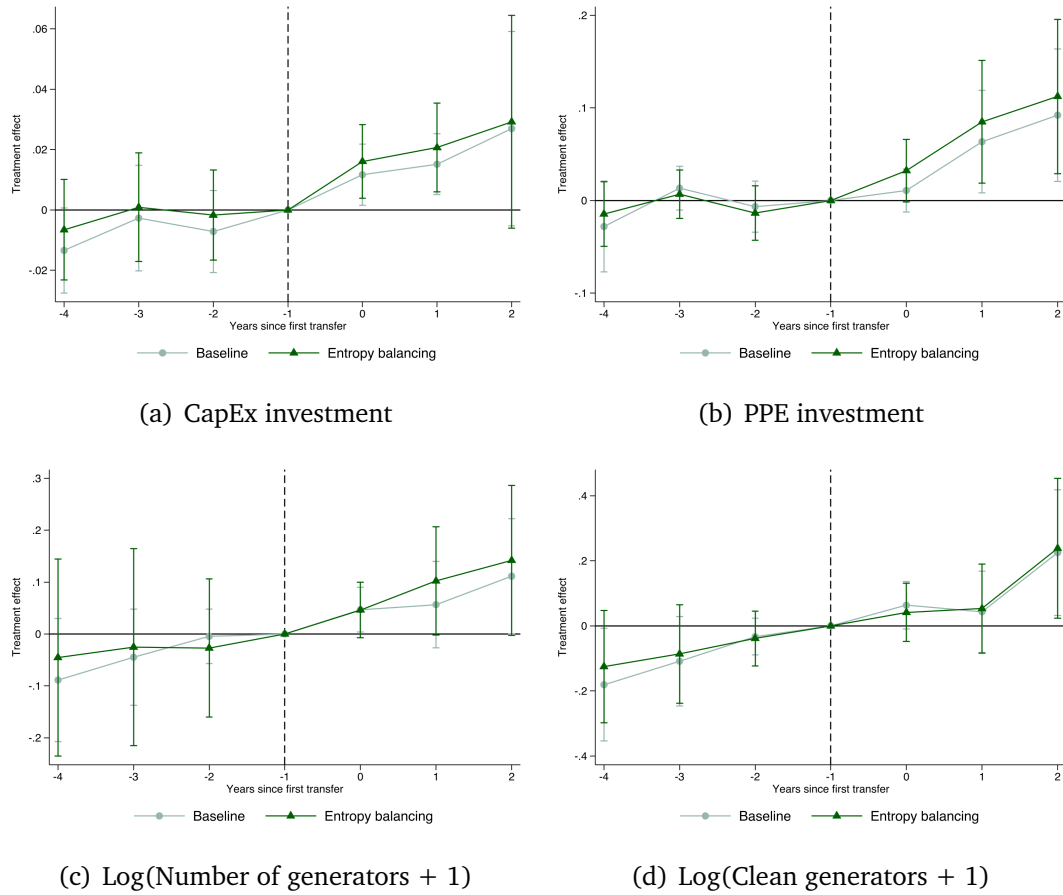
Notes: This figure plots event-study estimates of the effect of tax credit transfers on investment, using the alternative $\log(1 + y)$ transformation of each outcome: $\log(1 + \text{CapEx})$ in Panel (a) and $\log(1 + \text{PPE})$ in Panel (b), obtained by replacing the *Post* indicator with separate relative-year indicators. Both panels use Sample 1, the public electric utility firm-cohort-year panel, with cohort $\times$ firm and cohort $\times$ year fixed effects and one-year-lagged control variables. Each panel presents the baseline control group (turquoise circles) and entropy balancing (dark green triangles). The series are dashed at event years $t + 1$ and $t + 2$ because the 2024 and 2025 cohorts are only partially represented or not represented at these horizons. Capped vertical lines indicate 95% confidence intervals; standard errors are clustered at the firm level.

Table B.5: Seller-Side Responses – Robustness for Effect of Tax Credit Transfer on Generator Investment

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline			Entropy balanced		
Panel A:	Log(Generators + 1)					
Treat × Post	0.088** (2.58)	0.097** (2.18)	0.066** (2.51)	0.124*** (2.80)	0.113* (1.91)	0.048* (1.71)
Log(Net generation + 1) _{<i>t</i>−1}	0.022** (2.39)	0.032*** (2.67)	0.014*** (7.23)	0.044 (1.49)	0.105*** (3.34)	0.068*** (3.87)
Pct. clean generation _{<i>t</i>−1}	0.177 (1.14)	0.120 (0.68)	-0.061*** (-3.65)	0.144 (0.51)	0.288 (0.74)	-0.292* (-1.94)
Observations	2,479	1,087	58,344	2,479	1,087	58,344
Adj. R ²	0.984	0.982	0.983	0.977	0.970	0.990
Panel B:	Log(Clean generators + 1)					
Treat × Post	0.126** (2.22)	0.165** (2.05)	0.128*** (3.54)	0.139** (2.17)	0.151* (1.71)	0.078* (1.95)
Log(Net generation + 1) _{<i>t</i>−1}	-0.001 (-0.05)	0.003 (0.17)	0.008*** (3.87)	-0.010 (-0.50)	0.035* (1.76)	0.043** (2.21)
Pct. clean generation _{<i>t</i>−1}	0.491*** (2.65)	0.450** (2.37)	0.014 (0.52)	0.656*** (2.92)	1.155*** (5.02)	0.002 (0.01)
Observations	2,479	1,087	58,344	2,479	1,087	58,344
Adj. R ²	0.984	0.982	0.980	0.987	0.983	0.987
Panel C:	Log(Non-clean generators + 1)					
Treat × Post	-0.002 (-0.05)	0.033 (0.74)	0.000 (0.02)	0.030 (0.51)	0.063 (0.89)	0.005 (0.45)
Log(Net generation + 1) _{<i>t</i>−1}	0.028*** (2.89)	0.038*** (2.95)	0.007*** (4.30)	0.028 (0.72)	0.075 (1.62)	0.010*** (2.68)
Pct. clean generation _{<i>t</i>−1}	-0.126 (-0.68)	-0.194 (-0.93)	-0.083*** (-3.43)	0.031 (0.07)	-0.422 (-0.73)	-0.119** (-2.04)
Observations	2,479	1,087	58,344	2,479	1,087	58,344
Adj. R ²	0.981	0.979	0.992	0.976	0.969	0.998
Sample	Public	2023	+ Private	Public	2023	+ Private
Cohort × firm FE	X		X	X		X
Cohort × year FE	X		X	X		X
Firm FE		X			X	
Year FE		X			X	
Lagged controls	X	X	X	X	X	X

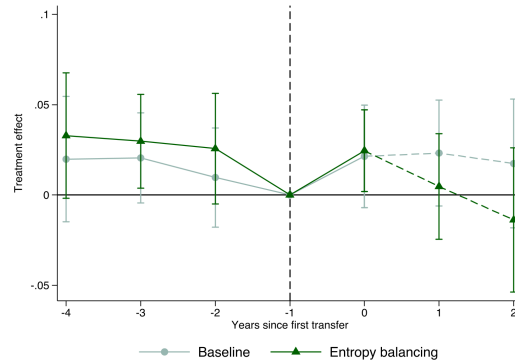
Notes: This table reports robustness checks for the effect of tax credit transfers on the number of generators, all as log(count + 1): overall generators (Panel A), and split into clean (Panel B) and non-clean (Panel C) generators. Columns (1) through (3) use the baseline control group, whereas columns (4) through (6) use entropy balancing. Within each set, the first column uses Sample 1, the public electric utility firm-cohort-year panel, with cohort × firm and cohort × year fixed effects and one-year-lagged controls; the second restricts Sample 1 to the 2023 cohort, the only cohort with a full post-treatment window, with lagged controls and (since a single cohort collapses cohort × firm and cohort × year fixed effects to plain firm and year fixed effects) firm and year fixed effects in place of the cohort-interacted terms; the third uses the same lagged-control specification as the first column but is estimated on Sample 2, the public and private electric utility firm-cohort-year panel. All variables are defined in Appendix B.1. t-statistics based on standard errors clustered at the firm level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure B.2: Seller-Side Responses – Effect of Tax Credit Transfer on Investment and Generators, 2023 Cohort, Event Studies

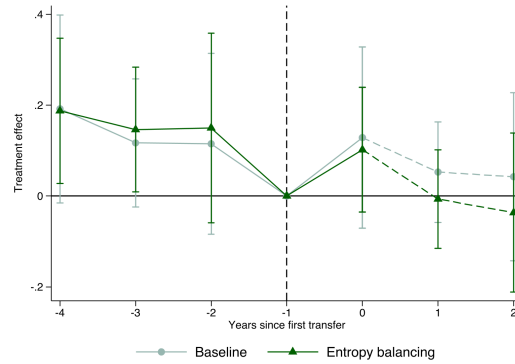


Notes: This figure plots event-study estimates of the effect of tax credit transfers on investment for the 2023 cohort, the only cohort with a full post-treatment window, obtained by replacing the *Post* indicator with separate relative-year indicators. The dependent variables are CapEx scaled by one-year-lagged total assets in Panel (a), PPE scaled by one-year-lagged total assets in Panel (b), log(Number of generators + 1) in Panel (c), and log(Number of clean generators + 1) in Panel (d). All panels use Sample 1, the public electric utility firm-year panel, with firm and year fixed effects and one-year-lagged control variables. Each panel presents the baseline control group (turquoise circles) and entropy balancing (dark green triangles). Capped vertical lines indicate 95% confidence intervals; standard errors are clustered at the firm level.

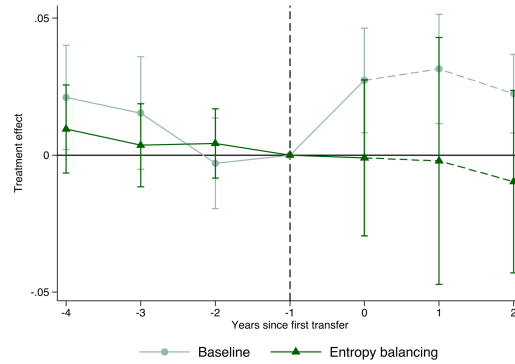
Figure B.3: Seller-Side Responses – Effect of Tax Credit Transfer on Investment Quality, Event Studies



(a) Capacity factor



(b) Log(Average generation + 1)



(c) ROA

Notes: This figure plots event-study estimates of the effect of tax credit transfers on investment efficiency: the capacity factor (Panel (a)), log(Average generation + 1) (Panel (b)), and log(ROA + 1) (Panel (c)), obtained by replacing the *Post* indicator with separate relative-year indicators. All panels use Sample 1, the public electric utility firm-cohort-year panel, with cohort $\times$ firm and cohort $\times$ year fixed effects and one-year-lagged control variables. Each panel presents the baseline control group (turquoise circles) and entropy balancing (dark green triangles). The series are dashed at event years $t + 1$ and $t + 2$ because the 2024 and 2025 cohorts are only partially represented or not represented at these horizons. Capped vertical lines indicate 95% confidence intervals; standard errors are clustered at the firm level.

Table B.6: Seller-Side Responses – Cross-Sectional Tests of Investment Quality Channels

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Entropy balanced									
Panel A:	<i>Monetization-Constraint Channel</i>									
	ROA		NOL/assets		Pretax income		Ever tax equity		Rate regulated	
	Low	High	High	Low	Loss	Profit	No	Yes	No	Yes
<i>Capacity factor</i>										
Treat × Post	-0.018 (-1.04)	-0.020* (-1.88)	0.043 (1.54)	0.096 (1.57)	0.006 (0.14)	-0.018 (-1.59)	-0.010 (-0.76)	0.007 (0.69)	-0.004 (-0.20)	-0.005 (-0.51)
Observations	823	1,656	315	623	200	2,279	2,377	102	1,502	935
Adj. R ²	0.918	0.889	0.964	0.879	0.926	0.868	0.906	0.952	0.929	0.893
<i>Log(Average generation + 1)</i>										
Treat × Post	-0.125* (-1.70)	-0.141** (-2.53)	-0.150** (-2.24)	1.243 (1.40)	0.137 (0.64)	-0.110** (-2.02)	-0.072 (-1.14)	-0.183 (-1.82)	-0.082 (-0.72)	-0.065 (-1.27)
Observations	823	1,656	315	623	200	2,279	2,377	102	1,502	935
Adj. R ²	0.980	0.965	0.991	0.976	0.978	0.962	0.974	0.940	0.964	0.972
<i>Log(ROA + 1)</i>										
Treat × Post	-0.022 (-1.10)	0.002 (0.69)	-0.043** (-2.11)	-0.033 (-0.90)	-0.032* (-1.78)	0.000 (0.16)	-0.007 (-0.49)	-0.011 (-0.84)	-0.009 (-0.32)	-0.001 (-0.32)
Observations	823	1,656	315	623	200	2,279	2,377	102	1,502	935
Adj. R ²	0.735	0.300	0.701	0.387	0.666	0.341	0.765	0.631	0.749	0.399
Sample	Public	Public	Public	Public	Public	Public	Public	Public	Public	Public
Cohort × firm FE	X	X	X	X	X	X	X	X	X	X
Cohort × year FE	X	X	X	X	X	X	X	X	X	X
Lagged controls	X	X	X	X	X	X	X	X	X	X

Table B.6: Seller-Side Responses – Cross-Sectional Tests of Investment Quality Channels (continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Entropy balanced									
Panel B:	<i>Financing-Friction Channel</i>									
	Firm size		Firm age		Leverage		HP index		Oper. CF	
	Small	Large	Young	Old	High	Low	High	Low	Neg.	Pos.
<i>Capacity factor</i> Treat × Post	0.004 (0.18)	-0.016 (-1.15)	-0.027 (-1.47)	-0.005 (-0.40)	-0.004 (-0.29)	-0.013 (-0.75)	-0.025 (-1.24)	-0.008 (-0.58)	0.020 (0.66)	-0.015 (-1.18)
Observations	829	1,647	842	1,634	821	1,658	823	1,656	107	2,364
Adj. R ²	0.931	0.867	0.938	0.881	0.947	0.883	0.940	0.881	0.960	0.879
<i>Log(Average generation + 1)</i> Treat × Post	-0.038 (-0.61)	-0.088 (-1.38)	-0.050 (-0.43)	-0.085 (-1.59)	-0.123* (-1.72)	-0.048 (-0.71)	-0.030 (-0.24)	-0.095* (-1.78)	-0.001 (-0.01)	-0.077 (-1.38)
Observations	829	1,647	842	1,634	821	1,658	823	1,656	107	2,364
Adj. R ²	0.980	0.947	0.965	0.973	0.993	0.960	0.964	0.973	0.898	0.956
<i>Log(ROA + 1)</i> Treat × Post	-0.041 (-1.47)	0.008 (1.38)	-0.021 (-0.66)	0.004 (1.08)	-0.035 (-1.29)	0.008 (1.15)	-0.024 (-0.73)	0.002 (0.51)	-0.062** (-2.26)	0.006 (1.07)
Observations	829	1,647	842	1,634	821	1,658	823	1,656	107	2,364
Adj. R ²	0.790	0.316	0.715	0.836	0.743	0.808	0.722	0.805	0.797	0.360
Sample	Public	Public	Public	Public	Public	Public	Public	Public	Public	Public
Cohort × firm FE	X	X	X	X	X	X	X	X	X	X
Cohort × year FE	X	X	X	X	X	X	X	X	X	X
Lagged controls	X	X	X	X	X	X	X	X	X	X

Table B.6: Seller-Side Responses – Cross-Sectional Tests of Investment Quality Channels (continued)

	(1)	(2)	(3)	(4)
Entropy balanced				
Panel C:	<i>Clean-Investment-Opportunity Channel</i>			
	Pre-clean gen		Pre technologies	
	Low	High	Few	Multiple
<i>Capacity factor</i>				
Treat × Post	0.018*	-0.059***	-0.023	-0.004
	(1.67)	(-2.78)	(-0.62)	(-0.44)
Observations	1,562	917	1,135	1,344
Adj. R ²	0.937	0.878	0.913	0.877
<i>Log(Average generation + 1)</i>				
Treat × Post	-0.017	-0.141	-0.105	-0.074
	(-0.32)	(-1.11)	(-0.64)	(-1.51)
Observations	1,562	917	1,135	1,344
Adj. R ²	0.983	0.962	0.954	0.978
<i>Log(ROA + 1)</i>				
Treat × Post	-0.011	0.005	-0.026	0.007
	(-0.73)	(1.02)	(-0.92)	(1.26)
Observations	1,562	917	1,135	1,344
Adj. R ²	0.818	0.674	0.757	0.316
Sample	Public	Public	Public	Public
Cohort × firm FE	X	X	X	X
Cohort × year FE	X	X	X	X
Lagged controls	X	X	X	X

Notes: This table reports the Treat × Post coefficient from separate subsample regressions testing the same three channels as Table 5, now on investment-quality outcomes: Capacity factor and log(Average generation + 1) use the generation controls (L_ingen/L_cleanshare); log(ROA + 1) uses the financial controls (L_inat/L_piat/L_leverage). Panel A measures the monetization-constraint channel using tax-capacity measures (ROA, NOL/assets, Pretax income) and firm characteristics (Ever tax equity, Rate regulated). Panel B measures the financing-friction channel using a financial-constraints index (HP index), firm characteristics (Firm size, Firm age, Leverage), and a cash-flow constraint (Oper. CF). Panel C measures the clean-investment-opportunity channel using each firm's pre-treatment clean generation share (Pre-clean gen, split at 0.5) and the number of distinct clean technologies deployed (Pre technologies, split at the sample median). ROA, NOL/assets, Pretax income, Leverage, Pre-clean gen, and Pre technologies are each firm's pre-treatment mean; Firm size, Firm age, HP index, and Oper. CF are measured in event year $t - 1$; all tercile and median cutoffs are computed once on the baseline (non-entropy-balanced) sample and applied identically to the entropy-balanced sample. Each column omits whichever control mechanically overlaps with that column's own split variable, applied separately to whichever of the two control sets (generation or financial) a given outcome uses. All variables are defined in Appendix B.1. Standard errors are clustered at the firm level; t-statistics are reported in parentheses beneath the coefficients. Significance levels are based on two-sided t-tests: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Buyer-Side Responses

C.1 Variable Definition Buyer-Side Responses

Table C.1: Buyer-Side Responses – Variable Definitions

Variable	Definition	Data source
Tax equity, $ever_i$	Indicator equal to one if firm i ever invested in tax equity over the sample period, and zero otherwise.	Hand-collected
Total assets $_{i,t}$	Total assets for firm i in year t , in \$ millions, winsorized at the 1st/99th percentile.	Compustat (at)
Firm age $_{i,t}$	Number of years firm i has appeared on Compustat, winsorized at the 1st/99th percentile.	Compustat (age)
Pretax income / assets $_{i,t}$	Pretax income divided by total assets for firm i in year t .	Compustat (pi/at)
ROA $_{i,t}$	Net income divided by total assets for firm i in year t .	Compustat (ni/at)
Leverage $_{i,t}$	Long-term debt divided by total assets for firm i in year t .	Compustat (dltt/at)
Cash / assets $_{i,t}$	Cash and short-term investments divided by total assets for firm i in year t .	Compustat (che/at)
GAAP ETR $_{i,t}$	Total income tax expense divided by pretax income net of special items for firm i in year t , winsorized at 0 and 1.	Compustat (txt/(pi-spi))
Cash ETR $_{i,t}$	Income taxes paid in cash divided by pretax income net of special items for firm i in year t , winsorized at 0 and 1.	Compustat (txpd/(pi-spi))
Federal GAAP ETR $_{i,t}$	Federal income tax expense divided by pretax income net of special items for firm i in year t , winsorized at 0 and 1.	Compustat (txfed/(pi-spi))
Federal taxes $_{i,t}$	Federal income tax expense for firm i in year t , in \$ millions, winsorized at the 1st/99th percentile.	Compustat (txfed)
Current ETR $_{i,t}$	Current-tax component of the GAAP ETR (total income tax expense less deferred income taxes) divided by pretax income net of special items for firm i in year t , winsorized at 0 and 1.	Compustat ((txt-txdi)/(pi-spi))
Deferred ETR $_{i,t}$	Deferred income taxes divided by pretax income net of special items for firm i in year t , winsorized at 0 and 1.	Compustat (txdi/(pi-spi))
NOL $_{i,t}$	Tax loss carryforward divided by total assets for firm i in year t (missing carryforward treated as zero).	Compustat (tlcf/at)
ESG score $_{i,t}$	Combined ESG score for firm i in year t , 0–100 scale, winsorized at the 1st/99th percentile.	LSEG (Refinitiv)

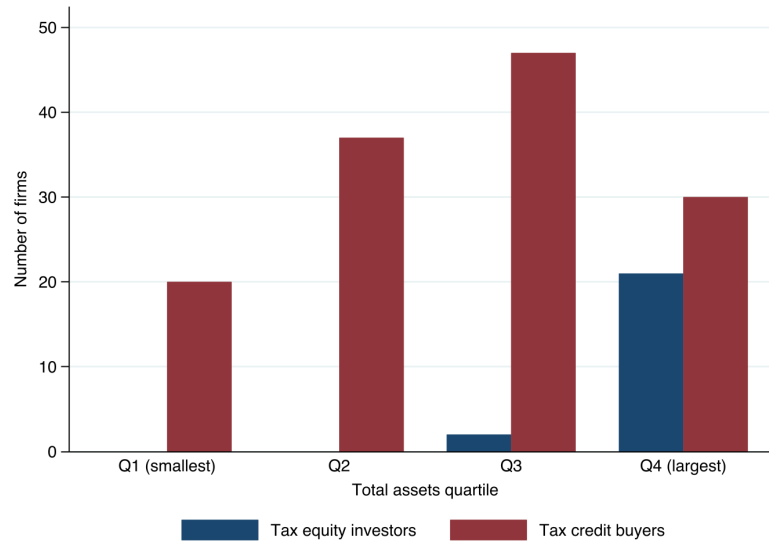
C.2 Additional Tables and Figures Buyer-Side Responses

Table C.2: Tax Equity Investment and Tax Credit Purchases by Year

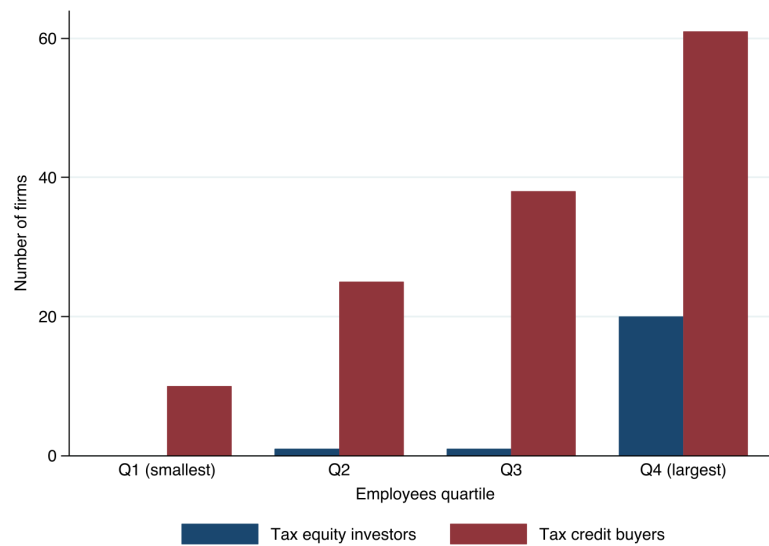
Year	Tax Equity Investment	Tax Credit Purchase
2015	6	0
2016	11	0
2017	9	0
2018	7	0
2019	6	0
2020	12	0
2021	7	0
2022	5	0
2023	9	14
2024	6	66
2025	4	126
Total	82	206

Notes: The table reports the number of firms with tax equity investments and tax credit purchases per year. A firm is counted in a given year if it invests in tax equity or purchases a tax credit in that year. The sample period is 2015 to 2025.

Figure C.1: Market Participation by Firm Size: Size Quartiles



(a) Total assets



(b) Employees

Notes: The figure shows the number of tax equity investors (navy) and tax credit buyers (maroon) by firm-size quartile, based on the firm's pre-treatment (2019–2022) mean size. Panel (a) uses total assets and Panel (b) uses number of employees to construct quartiles of firm size.

Table C.3: Buyer-Side Responses – Treated vs. Control: Stacked Cohort Panel

	Treated				Control				Diff.
	N	Mean	SD	Median	N	Mean	SD	Median	
Baseline									
Tax equity, ever	534	0.02	0.15	0.00	5,802	0.05	0.21	0.00	-0.02***
Total assets	534	71,160.60	275,785.45	10,808.80	5,802	112,569.69	312,892.64	6,759.87	-41,409.10***
Firm age	534	31.53	21.50	29.00	5,802	24.16	15.24	22.00	7.37***
Pretax income / assets	534	0.08	0.08	0.06	5,802	0.03	0.06	0.01	0.06***
ROA	534	0.06	0.07	0.05	5,802	0.02	0.06	0.01	0.05***
Leverage	534	0.27	0.27	0.23	5,802	0.09	0.15	0.03	0.18***
Cash / assets	534	0.10	0.12	0.06	5,802	0.11	0.12	0.07	-0.01**
GAAP ETR	519	0.21	0.11	0.21	5,711	0.20	0.12	0.20	0.01
Cash ETR	511	0.20	0.14	0.20	5,706	0.19	0.16	0.18	0.01
Federal GAAP ETR	399	0.14	0.12	0.15	1,459	0.12	0.15	0.10	0.02**
Federal Taxes	412	278.71	489.90	115.38	1,504	220.27	492.82	21.74	58.45**
Current ETR	415	0.21	0.15	0.21	2,256	0.18	0.18	0.17	0.03***
Deferred ETR	415	0.05	0.13	0.00	2,256	0.08	0.18	0.01	-0.03***
NOL	534	0.03	0.15	0.00	5,802	0.02	0.18	0.00	0.00
ESG score	481	51.99	19.44	54.00	3,748	41.47	17.95	39.15	10.52***
Emissions reduction target	506	0.30	0.46	0.00	4,423	0.11	0.32	0.00	0.19***
Entropy balancing									
Tax equity, ever	534	0.02	0.15	0.00	5,802	0.07	0.25	0.00	-0.05***
Total assets	534	71,160.60	275,785.45	10,808.80	5,802	143,569.48	361,632.14	7,489.00	-72,408.88***
Firm age	534	31.53	21.50	29.00	5,802	25.00	16.61	22.00	6.53***
Pretax income / assets	534	0.08	0.08	0.06	5,802	0.08	0.11	0.03	-0.00
ROA	534	0.06	0.07	0.05	5,802	0.06	0.11	0.02	-0.00
Leverage	534	0.27	0.27	0.23	5,802	0.27	0.33	0.11	-0.00
Cash / assets	534	0.10	0.12	0.06	5,802	0.13	0.12	0.09	-0.03***
GAAP ETR	519	0.21	0.11	0.21	5,711	0.20	0.11	0.21	0.01
Cash ETR	511	0.20	0.14	0.20	5,706	0.20	0.16	0.18	0.00
Federal GAAP ETR	399	0.14	0.12	0.15	1,459	0.14	0.11	0.13	0.00
Federal Taxes	412	278.71	489.90	115.38	1,504	198.02	439.09	42.45	80.70***
Current ETR	415	0.21	0.15	0.21	2,256	0.20	0.14	0.19	0.01
Deferred ETR	415	0.05	0.13	0.00	2,256	0.05	0.14	0.00	-0.00
NOL	534	0.03	0.15	0.00	5,802	0.02	0.19	0.00	0.00
ESG score	481	51.99	19.44	54.00	3,748	46.11	17.89	44.53	5.88***
Emissions reduction target	506	0.30	0.46	0.00	4,423	0.20	0.40	0.00	0.10***

Notes: The table reports summary statistics for treated and control firms in the pre period of the stacked cohort panel, under a baseline (unweighted) and an entropy-balanced (weighted) specification. Diff. is the treated-minus-control difference. *** p<0.01, ** p<0.05, * p<0.1.